Cardiovascular Critical Care II

Scott T. Benken, Pharm.D., MHPE, FCCP, FCCM, BCPS-AQ Cardiology

University of Illinois at Chicago, College of Pharmacy

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Learning Objectives

1. Manage cardiac arrest from the initiation of basic life support to the use of post–cardiac arrest care.
2. List the indications and special considerations for medication administration during cardiac arrest.
3. Categorize the patient groups that should receive temperature control after cardiac arrest.
4. Develop treatment plans for common complications of temperature control after cardiac arrest.
5. Analyze the therapeutic goals and clinical indications for the medications used in hypertensive emergency.

Abbreviations in This Chapter

ACLS	Advanced cardiac life support
AED	Automated external defibrillator
BLS	Basic life support
BPV	Blood pressure variability
CPR	Cardiopulmonary resuscitation
DBP	Diastolic blood pressure
ED	Emergency department
ICP	Intracranial pressure
ICU	Intensive care unit
IO	Intraosseous
MAP	Mean arterial pressure
MICU	Medical intensive care unit
PEA	Pulseless electrical activity
pVT	Pulseless ventricular tachycardia
ROSC	Return of spontaneous circulation
SCA	Sudden cardiac arrest
SBP	Systolic blood pressure
VF	Ventricular fibrillation
VT	Ventricular tachycardia

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

Questions 1–5 pertain to the following case.

T.B. is a 72-year-old man with a history of atrial fibrillation, coronary artery disease with drug-eluting stent placement in 2009, heart failure with reduced ejection fraction (most recent ejection fraction was 45% on echocardiogram in 2018), and gastroesophageal reflux disease. T.B. is sent to the catheterization laboratory for suspected acute myocardial infarction. Laboratory values for T.B. are as follows: international normalized ratio (INR) 1, platelet count 200,000/mm^3, hemoglobin 12 g/dL, serum creatinine (SCr) 1.7 mg/dL (baseline 1.5 mg/dL), white blood cell count (WBC) 17 x 10^3 cells/mm^3, and aspartate aminotransferase (AST) 100 IU/L. He is admitted to the coronary care unit for observation after catheterization when he suddenly loses consciousness and becomes pulseless. The coronary care unit team of which you are part is called to the bedside. Of note, T.B. has peripheral intravenous access, and before this event, he was on room air by nasal cannula.

1. Which is the most appropriate first step in T.B.'s resuscitation?
   A. Promptly intubate because this is likely a hypoxic pulmonary arrest.
   B. Place central line for vasopressor administration.
   C. Provide chest compressions.
   D. Give 2 breaths by bag-mask ventilator.

2. The monitor reveals that T.B. is in ventricular fibrillation (VF), and T.B. remains pulseless. Which is the most appropriate management of T.B.'s VF arrest?
   A. Provide an unsynchronized shock/defibrillation at 120 J from a biphasic defibrillator.
   B. Give intravenous amiodarone at a dose of 300 mg.
   C. Give intravenous atropine at a dose of 1 mg.
   D. Provide emergency transcutaneous pacing.

3. T.B.'s rhythm changes from VF to pulseless electrical activity (PEA) on the monitor. Which is the most appropriate management of T.B.'s PEA arrest?
   A. Provide an unsynchronized shock/defibrillation at 120 J from a biphasic defibrillator.
   B. Chest compressions while considering treatable causes of cardiac arrest.
   C. Give intravenous lidocaine 1 mg/kg x 1, followed by an infusion at 1 mg/minute.
   D. Give intravenous atropine at a dose of 1 mg.

4. T.B. has return of spontaneous circulation (ROSC) after 15 minutes of total resuscitation. The team is deciding whether temperature control would be appropriate for T.B. Which is the most accurate statement regarding temperature control for T.B.?
   A. He should not be considered because his initial presentation was VF arrest.
   B. He should not be considered because he has transaminitis.
   C. He should be considered with SCr monitoring.
   D. He should be considered, but thrombolysis should be initiated concurrently because of the associated risk of clotting.

5. The medical team wants further information about the literature regarding temperature control. Which is the most appropriate information regarding the data to support temperature control?
   A. Improves survival in PEA cardiac arrests.
   B. Shown to improve neurologic outcomes.
   C. Superiority found when targeting a core temperature of 36°C compared with 33°C.
   D. Most studied in patients with asystole.

Questions 6–8 pertain to the following case.

J.H. is a 48-year-old woman with no known medical history who presents to the emergency department (ED) for acute onset of shortness of breath, right-side pain, and some blurry vision. She denies any illicit drug or cigarette use, but she confirms social alcohol intake (about 3 drinks per week). Urine toxicology is negative. Initial vital signs are as follows: blood pressure (BP) 202/140 mm Hg, heart rate (HR) 88 beats/minute, respiratory rate (RR) 22 breaths/minute, and pain 4/10 (chest and right-side pain). Initial laboratory values are as follows: SCr 0.8 mg/dL, AST 608 U/L, ALT 458 U/L, lipase 20 U/L, total bilirubin (Tbil) 1 mg/dL, direct bilirubin (Dbil) 0.4 mg/dL, WBC 6 x 10^3 cells/mm^3, hemoglobin 11 mg/dL, troponin T less than 0.01 ng/mL, and D-dimer less than 0.5 mcg/mL. Chest radiography shows moderate bilateral pleural effusions and no focal consolidations. Chest computed tomography (CT) is negative for pulmonary embolism. Of note, J.H. is taking no prescription or over-the-counter medications.

6. Which laboratory abnormality or presenting symptom best qualifies J.H. for hypertensive emergency?
   A. None; she is having a hypertensive urgency.
   B. Her SCr.
   C. Her AST/ALT.
   D. Her D-dimer.

7. Which is the most appropriate initial treatment strategy for J.H.'s BP?
   A. Phentolamine 1 mg intravenously every 30 minutes.
   B. Metoprolol 25 mg orally every 12 hours.
   C. Nitroprusside 0.25 mcg/kg/minute by continuous intravenous infusion.
   D. Enalaprilat 10 mg intravenously every 6 hours.

8. Which is the most appropriate goal for J.H.'s BP reduction?
   A. Goal mean arterial pressure (MAP) reduction of 25% during the first 60 minutes.
   B. Goal MAP reduction of 50% during the first 60 minutes.
   C. Goal MAP reduction of 25% during the first 24 hours.
   D. Goal MAP reduction of 50% during the first 24 hours.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 4, 6
   b. Task B: 1, 2, 5
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1-3, 5, 6
   b. Task B: 1, 2, 4
   c. Task C: 1-4
3. Domain 3: Professional Practice
   a. Task A: 4
   b. Task B: 1, 2, 4

I. ADVANCED ADULT CARDIAC LIFE SUPPORT

A. Background
   1. Foundation of advanced cardiac life support (ACLS) is effective and timely basic life support (BLS).
   2. Sudden cardiac arrest (SCA) continues to be a leading cause of death in many parts of the world.
   3. SCA can vary in etiology (noncardiac vs. cardiac), circumstances (unwitnessed vs. witnessed), and setting (in vs. out of hospital).
   4. Because of heterogeneity, action links that are denoted the "chain of survival" were developed for guidance (Circulation 2016;133:447-54). There is an increased recognition in the differences between arrests that occur in the hospital and those that do not, which has led to two distinct chains of survival:
      a. In-hospital cardiac arrest (IHCA) chain of survival
         i. Appropriate surveillance and early warning systems (prevention)
         ii. Activation of multidisciplinary response team
         iii. Early CPR that emphasizes chest compressions
         iv. Rapid defibrillation, if indicated
         v. Advanced life support and post-cardiac arrest care
         vi. Recovery
      b. Out-of-hospital cardiac (OHCA) arrest chain of survival
         i. Immediate recognition of SCA and activation of emergency response system
         ii. Early CPR that emphasizes chest compressions
         iii. Rapid defibrillation, if indicated
         iv. Basic and advanced emergency medical services
         v. Advanced life support and post-cardiac arrest care
         vi. Recovery
   5. Following the chain of survival effectively can improve survival (e.g., with out-of-hospital, witnessed VF arrest, survival rates can approach 50%) (Circulation 2006;114:2760-5). Because BLS and ACLS are often experienced as a team approach in the hospital setting, it is imperative that hospital providers be familiar with all aspects of BLS and ACLS in order to fulfill any role during the arrest situation.

B. Basic and advanced emergency medical services
   1. Immediate recognition of SCA and activation of emergency response system (IHCA and OHCA - chain of survival): Unresponsive patient or witnessed sudden collapse with absent or gasping abnormal breathing (Circulation 2016;133:447-54; Acad Emerg Med 2007;14:877-83).
      a. Ensure that the scene is safe.
      b. Check for response by tapping on shoulder and shouting at patient; simultaneously check for normal breathing.
      c. Activate emergency response system (e.g., calling 911), and follow instructions from trained dispatchers/responders (Ann Emerg Med 1993;22:354-65). In the IHCA setting, this would include the activation of the emergency response team.
   2. Early CPR. Follow the sequence of C-A-B (compressions-airway-breathing) (IHCA and OHCA - chain of survival).
      a. C - Chest compressions are an essential component of CPR.
         i. Not often provided by nonmedical individuals until professional emergency responders arrive (Circ Cardiovasc Qual Outcomes 2010;3:63-81).
         ii. Both an increase in intrathoracic pressure and a direct compression of the heart lead to perfusion and oxygen delivery to the brain and myocardium.
         iii. All patients with SCA should receive chest compressions (Acta Anaesthesiol Scand 2008; 52:914-9).

iv. Place patient on a hard surface, use backboard (unless it will cause interruptions in chest compressions, delay in initiation of CPR, or dislodgment of lines/tubing), and/or deflate air-filled mattresses (J Intensive Care Med 2009;24:195-9; Acta Anaesthesiol Scand 2007;51:747-50; Resuscitation 2004;61:55-61).

v. Perform high-quality chest compressions at a rate of 100–120 compressions per minute at a depth of 2–2.4 inches, and allow chest recoil after each compression. High-quality chest compressions optimize coronary perfusion, cardiac index, myocardial blood flow, and cerebral perfusion (Crit Care Med 2010;38:1141-6; Resuscitation 2006;71:137-45; Resuscitation 2006;71:341-51; Circulation 2015;132(suppl 2):S315-S367).

vi. Without the use of a compression feedback device, it may be difficult to judge compression rate and depth. In a randomized study, compression feedback devices were shown to increase adherence to CPR guidelines, increase CPR quality, increase rates of ROSC, and decrease injury (e.g., rib fractures) seen with CPR (Crit Care 2016;20:147:1-8).

vii. Actual number of chest compressions given per minute is a function of the compression rate and proportion of time without interruption. Goal is to minimize interruptions to chest compressions.
   (a) Increasing the number of compressions given per minute can increase the rate of ROSC, increase survival rates, and increase rates of neurologically-intact survival (JAMA 2008;299:1158-64; Circulation 2009;120:1241-7; Circulation 2005;111:428-34).
   (b) Rescuer fatigue is common and may lead to inadequate compression quality (Resuscitation 2009;80:918-4). It is recommended to change compressors every 2 minutes (or after five cycles of compressions at a rate of 30:2 compressions/ventilations) with no more than 10 seconds between changes (Resuscitation 2009;80:1015-8).
   (c) Pulse checks (including initial) should last no more than 10 seconds.
   (d) Duration of the single longest interruption to chest compressions (regardless of reason) is negatively associated with survival (Circulation 2015;132:1030-7), reemphasizing the need to minimize interruptions to chest compressions.
   (e) Compression-first (or "compression only" or "hands only") CPR decreases time until first compression (Resuscitation 2004;62:283-9) and for suspected OHCA is acceptable for nonmedical rescuers (Circulation 2015;132(suppl 2):S315-S367).
   (f) A Cochrane review found that bystander-administered chest compression-only CPR (supported by telephone instruction) increases the number of people who survive to hospital discharge (Cochrane Database Syst Rev 2017;3:CD010134). It is unclear how well neurologic function is preserved in this population.

viii. Mechanical chest compression devices have not been shown to be superior to conventional CPR. They can be considered when prolonged CPR is used, being mindful of interruptions in CPR during device deployment and removal (Circulation 2015;132(suppl 2):S315-S367). A Cochrane review confirmed that mechanical chest compressions are not superior to manual chest compressions (Cochrane Database Syst Rev 2018;8:CD007260). Furthermore, in shockable rhythms, mechanical chest compression devices may lead to longer times to first defibrillation and greater intervals between additional defibrillations (Resuscitation 2017;115:155-62).

ix. Data suggest that patients can have ROSC with meaningful neurological recovery even after prolonged (>25 min) pre-hospital resuscitation efforts (Circulation 2016;133:1386-96; Resuscitation 2016;105:45-51), so extended CPR efforts can be anticipated.

x. The use of extracorporeal CPR is emerging as a therapy for those who fail conventional CPR. The current data seems promising in benefiting neurologic outcomes but is limited by bias and quality (Circulation 2019;140:e881-e894; Circulation 2024;149:e254-73). Current recommendation is that extracorporeal CPR may be considered for selected patients as rescue therapy when conventional CPR efforts are failing in settings in which it can be successfully implemented by skilled providers.

xi. Partial pressure of end-tidal CO_2 ($PETCO_2$), coronary perfusion pressure (aortic diastolic pressure minus right atrial diastolic pressure), arterial relaxation pressure, regional cerebral oxygenation and central venous oxygen saturation ($ScvO_2$) may correlate with cardiac output and myocardial blood flow during CPR. Threshold values have been reported at which ROSC is rarely achieved, and an abrupt increase in any of these variables is a sensitive indicator of ROSC (Table 1 - Circulation 2015;132(suppl 2):S315-S367; Ann Emerg Med 1992;21:1094-101; JAMA 1990;263:1106-13; Am J Emerg Med 1985;3:11-4; J Emerg Med 2010;38:614-21; Crit Care 2008;12:R115; Crit Care Med 2016;44:1663-74).

Table 1. Useful Physiological Variables During CPR

Variable	ROSC Rarely Achieved
$PETCO_2$	< 10 mm Hg
Coronary perfusion pressure	< 15–20 mm Hg
Arterial relaxation (diastolic) pressure	< 20–40 mm Hg
Regional cerebral oxygenation	< 25%
$ScvO_2$	< 30%

Patient Case

1. A.C., a 50-year-old man with a history of gastroesophageal reflux disorder and chronic obstructive pulmonary disease, was admitted for shortness of breath, palpitations, and presumed exacerbation of his lung disease. On hospital admission day 4, A.C. has a witnessed cardiac arrest on the family medicine unit. The emergency response team of which you are part is called, and when you arrive, the bedside nurse has already begun chest compressions. Which insight would be best shared regarding chest compressions for A.C.?
   A. Because the nurse has already begun chest compressions, she should continue chest compressions for the duration of CPR.
   B. Compressions increase intrathoracic pressure and directly compress the heart, which can generate cardiac output and deliver oxygen.
   C. Because increasing the intrathoracic pressure is vital to oxygen delivery, chest recoil is unnecessary and should be avoided.
   D. Number of chest compressions per minute does not affect outcomes, so pulse check and line and airway placements can interrupt chest compressions.

b. A - Airway
   i. To place an artificial airway in patients without an advanced airway, use the head-tilt, chin-lift technique if patients have no evidence of head or neck trauma and the jaw thrust alone if cervical spine injury is suspected (JACEP 1976;5:588-90; JAMA 1960;172:812-5) (see "Rescue Breaths" following).
   ii. Cricoid pressure is the technique of applying pressure to the patient's cricoid cartilage to push the trachea posteriorly and compress the esophagus with the goal of preventing aspiration.
      (a) May help in visualizing vocal cords during tracheal intubation.
      (b) Recommend against use for adult cardiac arrest because of possible delay or prevention of advanced airway, lack of protection from aspiration, and lack of mastery from "expert" and nonexpert rescuers (Emerg Med Australas 2005;17:376-81; Br J Anaesth 1994;72:47-51).

iii. If a foreign body airway obstruction (FBAO) occurs:
  (a) Observe the patient with a mild FBAO. Mild FBAO is defined by a patient with a partial airway obstruction that is still able to talk and has an effective cough.
  (b) Signs of severe FBAO include a silent cough, stridor, or increasing respiratory difficulty. If these occur, ask the patient, "Are you choking?" If the patient clutches their neck (universal sign of choking) or nods without answering verbally, consider severe FBAO:
    (1) Activate the emergency response system.
    (2) Administer abdominal thrusts to nonobese adults.
    (3) In adults with obesity or women in the late stage of pregnancy, administer chest thrusts.
  (c) If the patient becomes unresponsive:
    (1) Place on ground and begin CPR as chest compressions have been shown to generate higher airway pressure than abdominal thrusts (Resuscitation 2000; 44:105-8).
    (2) Each time the airway is opened during CPR to provide a rescue breath, look for an object in the patient's mouth and, if found, remove it. If not found, continue giving the rescue breaths (two total breaths), followed by 30 chest compressions.
    (3) No studies have evaluated the routine use of the finger sweep to clear an airway in the absence of visible airway obstruction. Case reports have shown some efficacy, but harm has also been demonstrated in patients and rescuers. A finger sweep should not be used in the absence of visible airway obstruction.

c. B - Rescue breaths
  i. Primary purpose is to assist in maintaining oxygenation, with secondary purpose of eliminating carbon dioxide (CO_2).
  ii. Compressions should always be initiated first as the arterial oxygen content of blood remains unchanged until CPR is initiated.
  iii. Optimal compression/ventilation ratio, inspired oxygen concentration, tidal volume, and RR yet to be determined.
  iv. Compression-only CPR without rescue breaths is recommended for nonmedical rescuers for OHCA (see earlier text for more detail). Trained nonmedical rescuers are encouraged to perform rescue breaths at a ratio of 30 compressions to 2 breaths. When the patient has an advanced airway in place during CPR, rescuers no longer deliver cycles of 30 compressions and 2 breaths (i.e., they no longer interrupt compressions to deliver 2 breaths). Instead, the provider may reasonably deliver 1 breath every 6 seconds (10 breaths/minute) while continuous chest compressions are being performed (Circulation 2015;132(suppl 2):S414-35).
    (a) Low minute ventilation (low tidal volume and/or low RR) can maintain oxygenation and ventilation during CPR as there is low oxygen uptake at the tissues and low CO_2 production (Circulation 1997; 95:1635-41).
    (b) Give sufficient tidal volume to produce visible chest rise (Resuscitation 1996;31:231-4). Excessive ventilation can increase intrathoracic pressure and decrease venous return as well as cause gastric inflation, which can lead to aspiration, regurgitation, and decreased survival (Circulation 2004;109:1960-5; Resuscitation 1998;36:71-3; JAMA 1987; 257:512-5).
    (c) Continuous chest compressions with asynchronous ventilation compared with chest compressions with interruptions for synchronous ventilation have not improved outcomes when delivered by emergency medical services (N Engl J Med 2015;23:2203-14).
  v. Deliver each rescue breath over 1 second. Mouth-to-mouth, mouth-to-barrier, mouth-to-stoma, and mouth-to-nose variations in initial rescue breathing are all acceptable and can produce oxygenation and ventilation (Chest 1994;106:1806-10; Br J Anaesth 1964;36:542-9).

vi. Positive-pressure ventilation
  (a) Bag-mask ventilation
    (1) Components include a nonjam inlet valve, either no pressure relief valve or a pressure relief valve that can be bypassed, standard 15-mm/22-mm fittings, an oxygen reservoir to allow for high oxygen delivery, and a non-rebreathing outlet valve (Respir Care 1992;37:673-90; discussion 690-4).
    (2) Should not be used by a single rescuer.
    (3) Should use an adult bag (1 or 2 L) and deliver (two-thirds or one-third of bag volume, respectively) about 600 mL of tidal volume, which should produce chest rise, oxygenation, and normocarbia (Resuscitation 2005;64:321-5; Resuscitation 2000;43:195-9).
    (4) Quantitative waveform capnography with a bag-mask device to confirm and monitor CPR quality is now recommended. Prior to the 2020 guidelines, the recommendation for quantitative waveform capnography monitoring was limited to monitoring only after endotracheal tube placement (https://acls-algorithms.com/2021-aha-acls-guideline-changes/adult-cardiac-arrest-algorithm-changes/)
  (b) Supraglottic airway devices (e.g., King Airway device) are considered an acceptable alternative to bag-mask ventilation during cardiac arrest (assuming proper training is supplied to rescuer) (Circ J 2009;73:490-6; Prehosp Emerg Care 1997;1:1-10).
    (1) In OHCA, supraglottic airway devices may be associated with an increased rate of ROSC and faster time to airway placement, but survival outcomes are uncertain (Crit Care Med 2024;52:e89-99).
    (2) Supraglottic airway devices are considered within the scope of BLS in some districts.
  (c) See the ACLS section for more details and information regarding endotracheal intubation.

vii. Naloxone for opioid-related life-threatening emergencies: Now with an independent algorithm in the most recent guidelines (Circulation 2020;142(16_suppl_2):S366-468).
  (a) Should be considered when a pulse is present but the patient has abnormal breathing or gasping (e.g., respiratory arrest).
  (b) Can consider (in addition to BLS) administration of intramuscular or intranasal naloxone.
  (c) If patients lose their pulse, provision of CPR should commence with continued consideration of naloxone if opioid intoxication is suspected.
  (d) Administer 0.4 mg intramuscularly or 2 mg intranasally diluted in 3 mL of normal saline – may repeat every 4 minutes.

Patient Case

2. L.S. is a 66-year-old woman visiting her husband at the hospital on the hospice unit. She is buying lunch in the cafeteria, and while in line to check out, she collapses. The emergency response team of which you are part is summoned. L.S. does not respond to voice or tapping of the shoulder, and a brief look at her chest shows no chest movement. Chest compressions are initiated while the crash cart and defibrillator are retrieved. Of note, a bag-mask ventilator is available at the scene because it is carried with the emergency response team. Which is most accurate about L.S.'s airway and breathing management?

A. A compression/ventilation ratio of 15:2 should be used throughout resuscitation efforts.
B. Because it is a multiple-rescuer scene, bag-mask ventilation should not be used because it is recommended only in single-rescuer resuscitations.
C. Rescue breaths should be given at a ratio of 30 compressions to 2 breaths, avoiding excessive ventilation given as synchronous ventilations.
D. Bag-mask ventilation should not be used in any patient because advanced airways are preferred to supply oxygen and eliminate CO_2.

3. Rapid defibrillation with a manual or automated external defibrillator (AED) (IHCA and OHCA - chain of survival)
    a. Defibrillation shock = unsynchronized shock.
    b. Successful defibrillation is defined as 5 seconds or greater of termination of arrhythmia after a shock is delivered.
    c. Early defibrillation of VF is crucial because it is the most common rhythm in witnessed OHCA, survival diminishes rapidly over time, and VF often progresses to asystole (Resuscitation 2000;44:7-17; Circulation 1997;96:3308-13).
    d. Three key actions must occur within moments of VF SCA to increase the likelihood of survival: (1) activation of the emergency medical services system (e.g., emergency response team), (2) provision of CPR, and (3) shock delivery (Ann Emerg Med 1993;22:1652-8).
    e. Performing chest compressions while a defibrillator is obtained significantly improves the probability of survival (Circulation 2009;120:1241-7). When VF is present for more than a few minutes, the myocardium becomes depleted of oxygen and energy substrates (e.g., adenosine triphosphate [ATP]).
        i. CPR can provide the oxygen and ATP needed until the shock is delivered.
        ii. Increased likelihood of termination of VF from shock delivery and ROSC if CPR given first (Circulation 2004;110:10-5).
        iii. If CPR is initiated immediately, survival can double or triple at most time intervals until defibrillation occurs (Resuscitation 2000;44:7-17; Ann Emerg Med 1995;25:780-4; Ann Emerg Med 1993;22:1652-8).
    f. Early defibrillation is a powerful predicator of ROSC after VF.
        i. Survival rates are highest for VF when CPR and defibrillation occur within 3–5 minutes of the event (Circ Cardiovasc Qual Outcomes 2010;3:63-81; Resuscitation 2009;80:1253-8). In the IHCA setting, prompt defibrillation (less than 2 minutes from the VF event) was associated with higher rates of long-term survival (Circulation 2018;137:2041-51).
            (a) For every minute that passes after collapse, survival from VF decreases 7%–10% (Ann Emerg Med 1993;22:1652-8).
            (b) CPR prolongs VF and delays the progression to asystole (Resuscitation 2000;47:59-70; Am J Emerg Med 1985;3:114-9).
        ii. CPR should be initiated immediately, with shock delivery as soon as possible.
        iii. One-shock biphasic (bidirectional) shock protocols are better than or equivalent to three-shock monophasic (one-directional) stacked protocols in terminating VF.
            (a) Almost all AEDs manufactured currently are biphasic.
            (b) Polyphasic waveform defibrillators are currently under investigation. Double-sequential defibrillation is not supported (Circulation 2020;142:S366-S468).
        iv. With any shock delivery, chest compressions should resume immediately, and pulse check should be delayed until the end of the next cycle of CPR because this can increase successful defibrillation with ROSC (Circulation 2004;110:10-5; Circulation 2002;105:2270-3).
        v. The optimal shock energy for biphasic first shock has yet to be determined.
            (a) Low-energy dosages (200 J or less) are safe and have equivalent or higher efficacy of termination of VF than monophasic waveform shocks at the same or higher energy (Circulation 2007;115:1511-7; Prehosp Emerg Care 2000;4:305-13).
            (b) The energy dosage used should be according to the manufacturer's recommendation (e.g., 120 or 200 J).
        vi. If additional shocks are needed, it is recommended that at least equivalent and potentially higher energy be used. Double-sequential defibrillation – shock delivered by two defibrillators almost simultaneously – is a new approach, but a systematic review has failed to establish its usefulness in resuscitation (Circulation 2020;142(suppl 1):S92-S139).

vii. Electrode placement
(a) Pad positioning is equally effective in terminating ventricular arrhythmias in four positions: anterolateral, anteroposterior, anterior-left infrascapular, and anterior-right infrascapular (Medicina (Kaunas) 2006;42:994-8; Physiol Meas 2006;27:1009-22).
(b) Lateral pads should be placed under breast tissue, and hirsute men should be shaved before the placement of pads.
(c) In patients with an implantable cardioverter-defibrillator or a pacemaker, it may be beneficial to avoid placing the pads or paddles over the device.
(d) Do not place the pads on top of a medication patch because this can cause current impedance or burning of the skin (Am J Emerg Med 1992;10:128-9).
viii. In-hospital AED use should be considered in ambulatory and unmonitored areas.
g. Pulseless VT (pVT) is treated like VF.
h. Pacing is not recommended for unstable VF or pVT (Circulation 2010;122:S685-705).

Patient Case

3. T.V. is a 72-year-old man with a history of chronic liver disease, hypoglycemia, and atrial fibrillation. He was admitted to the medical intensive care unit (MICU) 2 days ago for sepsis requiring aggressive fluid resuscitation and intravenous antibiotics. T.V. did not require vasopressors to treat his sepsis. On ICU day 3, T.V. develops VF on telemetry, loses consciousness, and becomes pulseless; the MICU team is summoned for a presumed VF cardiac arrest. Pads are placed on T.V. by the time the team arrives, and the rhythm is confirmed to be VF. Which is the most accurate statement regarding defibrillation for T.V.?

A. Three vital actions with VF can lead to increased survival if they occur rapidly: activate emergency response system, provide CPR, and deliver shock.
B. T.V. should not be defibrillated but should be paced out of the VF, if possible, because pacing is more effective for pulseless VF.
C. Chest compressions should be delayed until the defibrillator is charged because defibrillation is the definitive treatment of VF.
D. Alternative treatments such as antiarrhythmics, vasopressors, and magnesium should be tried first.

C. ACLS Interventions
1. Airway control and ventilation
a. Background
i. During CPR, oxygen delivery to the heart and brain becomes more flow-dependent than arterial oxygen saturation–dependent (Ann Emerg Med 1990;19:1104-6).
ii. Placement of an advanced airway in cardiac arrest should not delay CPR or defibrillation.
iii. No studies address the optimal timing of advanced airway placement. The guiding general concept is to place the advanced airway while minimizing interruptions to chest compressions.
iv. Conflicting evidence exists for the urgent placement of an advanced airway.
(a) Study of IHCA has shown an increased 24-hour survival in patients with an advanced airway placed within 5 minutes. No difference was found regarding ROSC (Resuscitation 2010;81:182-6); however, another study has shown a worse overall survival rate in cardiac arrest patients who required intubation (Arch Intern Med 2001;161:1751-8).
(b) OHCA studies have shown that intubation in the rural and urban setting and, more specifically, intubation within 13 minutes, is associated with better survival (Med J Aust 2006;185:135-9; Prehosp Emerg Care 2004;8:394-9).

(c) One large randomized controlled trial comparing endotracheal intubation to bag-mask ventilation in OHCA found no difference in 28-day survival or 28-day survival with favorable neurological function (JAMA 2018;319:779-87).

b. Oxygen during CPR
   i. Unclear what the optimal concentration of inspired oxygen content should be during CPR, but it is currently recommended that maximum (e.g., 100%) inspired oxygen be used to optimize arterial oxyhemoglobin content and, subsequently, oxygen delivery (Circulation 2015;132(suppl 2):S315-S367).
   ii. In OHCA, patients with a higher Pao_2 had higher survival to hospital admission but not neurologically intact survival, which is likely a function of underlying pathophysiology (Resuscitation 2013;84:770-5).
   iii. Extended exposure to high oxygen concentrations carries the risk of toxicity, but this toxicity has not been shown in the short-term setting of adult CPR (Resuscitation 1999;42:221-9).
   iv. During chest compressions, air is forcefully expelled from the chest, and oxygen is drawn into the chest by passive recoil. Because the ventilation requirements are lower than normal, passive oxygen delivery is theorized to be sufficient for several minutes of initial CPR (Circulation 1994;90:3070-5), but recommendations to remove ventilation cannot be made.

c. Bag-mask ventilation: Viable option for oxygenation and ventilation during CPR but should be provided only when there is more than one rescuer and/or trained personnel (for more details, see earlier discussion) (Circulation 2010;122:S729-767).

d. Airway adjuncts
   i. *Cricoid pressure* should be used only in special circumstances to help visualize the vocal cords and should be relaxed, released, or adjusted if it impedes ventilation or advanced airway placement.
   ii. *Oropharyngeal airways* can be considered to help facilitate bag-mask ventilation in the unresponsive patient with no cough or gag reflex.
   iii. *Nasopharyngeal airways* can be considered in patients with airway obstruction and clenched jaw but should be used cautiously in craniofacial injury and avoided in known coagulopathy because of an increased risk of bleeding (J Trauma 2000;49:967-8; Anaesthesia 1993; 48:575-80).

e. Advanced airways
   i. Endotracheal intubation
      (a) Attempted placement by unskilled providers leads to unacceptably large periods of chest compression interruption and hypoxemia.
      (b) Benefits include keeping the airway patent, allowing for suctioning of airway, ensuring high oxygen concentration delivery, providing a third line medication administration option, allowing for specific tidal volume delivery, and providing protection from aspiration.
   ii. Supraglottic airways
      (a) Do not require visualization of glottis, which allow for continuous chest compressions.
      (b) Types studied during cardiac arrest include laryngeal mask airway, esophageal-tracheal tube (Combitube), laryngeal tube (laryngeal tube or King Airway LT), among others. No difference in successful prehospital ventilation, ROSC, or 1-month neurologic outcome between laryngeal mask airway and laryngeal tube for OHCA (Am J Emerg Med 2015;33:1360-3).
      (c) When used by trained providers, they allow as effective oxygenation and ventilation as bag-mask ventilation and endotracheal intubation.

iii. Recommendations regarding type of advanced airway are equivocal (Circulation 2019;140:e881-e894).
   (a) Supraglottic airways can be used for adults with OHCA in settings with low tracheal intubation success rate or minimal training opportunities for ETT placement.
   (b) Supraglottic airways or ETT can be used for adults with OHCA in settings with high tracheal intubation success rates or optimal training opportunities for ETT placement.
   (c) If an advanced airway is used in the in-hospital setting by expert providers trained in these procedures, either the supraglottic airway or ETT can be used.
   (d) Frequent experience/retraining is recommended for providers who perform endotracheal intubation.
   (e) Emergency medical services systems that perform prehospital intubation should provide a program of ongoing quality improvement.

iv. After advanced airways are secured, proper placement should be confirmed with clinical assessment and objective measures without interruptions to chest compressions.
   (a) Physical assessments include visually inspecting chest rise bilaterally and listening to the epigastrium (breath sounds should be absent) and lung fields (should be equal and adequate).
   (b) Exhaled CO_2 or esophageal detector devices are a reasonable and objective means of confirmation if continuous waveform capnography is not readily available.
   (c) Continuous waveform capnography is the most reliable and objective way to ensure, confirm, and monitor correct endotracheal tube placement. Although not specifically studied with supraglottic airways, readings should be similar to endotracheal readings.
   (d) False-positive CO_2 detection (CO_2 detected not from ventilation) is rare, whereas false-negative CO_2 detection (no CO_2 detection when ventilation is occurring) is more common. Most common cause of false-negative CO_2 detection is a reduction in blood flow or CO_2 delivery to lungs (e.g., lack of quality chest compressions, pulmonary embolism, severe airway obstruction). Partial pressure of end-tidal CO_2 ($PETCO_2$) less than 10 mm Hg during CPR suggests ROSC is unlikely, and maneuvers such as improving quality of chest compressions, adding vasopressor therapy, and others, should be considered.

v. Post-intubation airway management
   (a) Airway should be marked (from front of teeth/gums) and secured (with tape or commercial device), avoiding compression around the neck, which could impair venous return from brain.
   (b) Chest radiography is suggested for confirmation of location of end of endotracheal tube in relation to the carina.
   (c) Slower ventilator rates (6–12 breaths/minute) have been shown to improve hemodynamic values and short-term survival in animal models of cardiac arrest (Crit Care Med 2006;34:1444-9; Circulation 2004;109:1960-5; Resuscitation 2004;61:75-82).

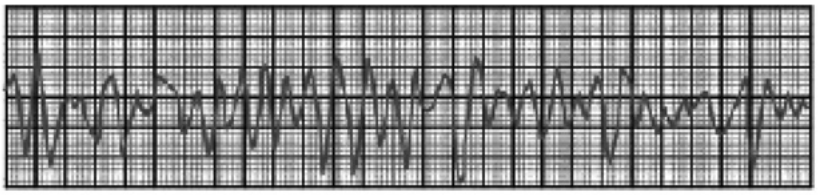
vi. After placement, continuous chest compressions should be given at a rate of 100-120 compressions per minute. A breath should be delivered every 6–8 seconds (8–10 breaths/minute), making sure to avoid over-ventilation, which could decrease venous return and cardiac output.

Patient Case

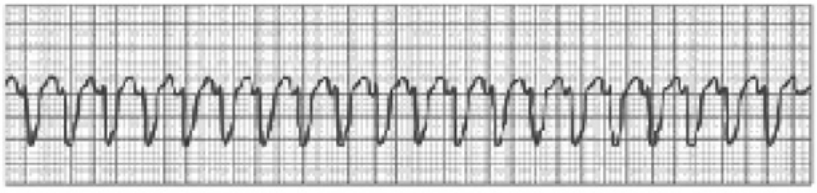
4. F.V. is a 63-year-old woman with a history of diabetes, heart failure with preserved ejection fraction, hypertension, and obstructive sleep apnea who presents to the ED with chest tightness and "feeling funny." In the ED, F.V. loses consciousness and develops pVT. Chest compressions are initiated immediately, pads are placed, and bag-mask ventilation is given at a compression/ventilation ratio of 30:2 for two cycles. The monitor confirms the rhythm of pVT. The defibrillator is charged, the patient is cleared, and the first shock is delivered. Chest compressions resume, and during the next pulse check, the patient is intubated. F.V. still has no pulse, and chest compressions are continued. Which is the most accurate statement about F.V.'s resuscitation after advanced airway placement?

   A. The compression/ventilation ratio should remain 30:2 to avoid excessive ventilation.
   B. The advanced airway should have been placed before defibrillation and CPR for pVT.
   C. The patient should be placed on room air (21% F_{IO_2} [fraction of inspired oxygen]).
   D. Proper airway placement should be confirmed with clinical assessment and objective measures.

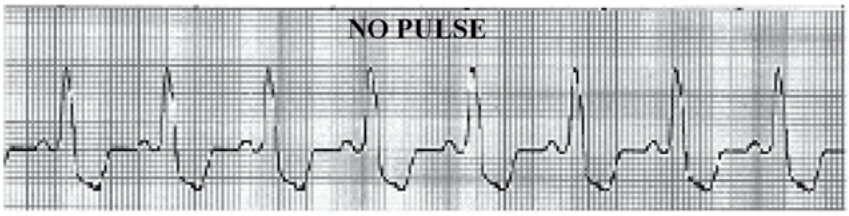
2. Management of cardiac arrest
   a. Background
      i. Cardiac arrest can be caused by four primary "rhythms": VF, pVT, PEA, and asystole.
      ii. These rhythms can also be classified as shockable (VF and pVT) and non-shockable (PEA and asystole).
         (a) Ventricular fibrillation
            (1) Wide complex
            (2) Polymorphic
            (3) Disorganized
            (4) Coarse or fine
            (5) No/minimal forward flow



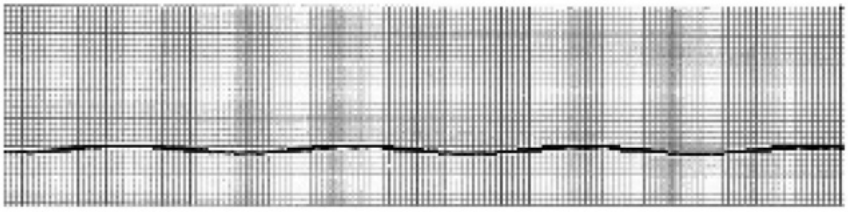
         (b) Pulseless VT
            (1) Wide complex
            (2) Monomorphic or polymorphic
            (3) Generally organized
            (4) No/minimal forward flow



(c) Pulseless electrical activity
  (1) Not a rhythm itself but defined as an organized rhythm that would be expected to produce mechanical activity but does not
  (2) Absent or insufficient mechanical ventricular activity
  (3) No/minimal forward flow



(d) Asystole (ventricular asystole)
  (1) Absence of detectable ventricular electrical activity
  (2) Accompanied by absence of ventricular mechanical activity
  (3) No/minimal forward flow



iii. Survival requires both BLS and ACLS. Foundation of ACLS is high-quality BLS. In addition to CPR, the only proven rhythm-specific therapy that increases survival at hospital discharge is defibrillation of VF/pVT.

iv. Medications have not consistently been shown to increase survival of cardiac arrest but have been shown to increase the rate of ROSC or survival to hospital admission (Resuscitation 2010;81:182-6; Med J Aust 2006;185:135-9; Resuscitation 2000;45:161-6; N Engl J Med 1999;341:871-8).
  (a) Vascular access and medication delivery should never interrupt CPR and/or defibrillation. All other therapies are "considerations" and should never compromise chest compressions.
  (b) If interruptions in chest compressions are necessary (e.g., rhythm assessment, delivery of defibrillation shocking in VF/VT), recommendation is to minimize duration (less than 10 seconds).

v. During cardiac arrest treatment, it is imperative to evaluate, treat, and/or reverse any treatable causes of cardiac arrest (Table 2).

vi. Post–cardiac care should begin immediately after ROSC is obtained to avoid re-arrest.

Table 2. Treatable Causes of Cardiac Arrest

H's	T's
Hypoxia	Toxins
Hypovolemia	Tamponade (cardiac)
Hydrogen ion (acidosis)	Thrombosis • Pulmonary embolism • Coronary thrombosis
Hypoglycemia	
Hypo/hyperkalemia	Tension pneumothorax
Hypothermia	

Patient Case

5. V.B., a 62-year-old man with an unknown medical history, comes to your ED altered and incoherent. He is admitted to the ED for observation, where he suddenly becomes unconscious and pulseless. The ED staff immediately initiates CPR for V.B., who is found to be in PEA. Which statement best describes appropriate cardiac arrest treatment?
   A. Survival from PEA is solely dependent on medications and advanced airways, so airway intubation and line placement take priority over CPR.
   B. The treatable causes of PEA cardiac arrest should be reviewed and addressed while CPR and laboratory tests are sent.
   C. Because the patient is in PEA, pads should be placed, the patient cleared, and shock delivered immediately.
   D. No strategies exist for post-arrest care to avoid re-arrest or improve outcomes from the PEA cardiac arrest.

b. Medication background (rhythm independent considerations)
   i. Goal: Increase myocardial blood flow during CPR and help achieve ROSC
   ii. Drug delivery
      (a) Central intravenous administration is recommended, if available. Central line placement should not interrupt CPR. The advantage of central administration is higher peak concentration and shorter drug circulation times than with peripheral routes (Crit Care Med 1988;16:1138-41; Ann Emerg Med 1981;10:73-8; Ann Emerg Med 1981;10:417-9).
      (b) If peripheral administration of medications is necessitated, a bolus injection should be followed by a 20-mL bolus of an intravenous fluid (e.g., 0.9% sodium chloride) to facilitate drug flow from the extremity to the central circulation (Am J Emerg Med 1990;8:190-3).
      (c) If proximal humerus intraosseous (IO) access is used, administration and drug delivery are similar to central venous access. If proximal or distal tibial intraosseous access is used, administration and drug delivery are similar to peripheral venous access and would necessitate a 20-mL bolus or pressure bag to facilitate drug administration. In a randomized cadaver study, tibial IO access has been shown to be easier to obtain, which allows for faster medication administration (Medicine 2016; 95: e3724). Data analyses suggest that intraosseous access is associated with poorer OHCA survival and fewer favorable neurologic outcomes, even after controlling for many other variables (Ann Emerg Med 2018;71:588-96). Intraosseous access may have initially been selected by first responders because of perceived complexity and influential patient characteristics. However, additional analysis has found that tibial intraosseous access compared with intravenous access is associated with a lower likelihood of ROSC and survival to hospitalization (Resuscitation 2017;117:91-6). The most recent guidelines now support the use of intravenous access preferentially unless it cannot be obtained (Circulation 2020;142:S366-S468). In these circumstances, intraosseous access is reasonable.
      (d) Endotracheal (ET) delivery
         (1) NAVEL acronym summarizes medications that have been studied and shown to have tracheal absorption by ET tube delivery. N – naloxone, A – atropine, V – vasopressin, E – epinephrine, L – lidocaine.
         (2) Serum concentrations of medications are lower when given by ET delivery.
         (3) Optimal ET doses unknown; typically, they are 2–2.5 times intravenous doses but may be higher (e.g., up to 3–10 times higher for epinephrine) (Crit Care Med 1987;15:1037-9).
         (4) Should be diluted with either 5–10 mL of 0.9% sodium chloride or sterile water and injected directly into the ET tube, ideally during pulse check to avoid medication expulsion back out the ET (Crit Care Med 1994;22:1174-80)

(e) Peak effect of intravenous or intraosseous medication delayed 1–2 minutes during CPR.
(f) Theoretical concern that giving high-dose bolus vasopressors after ROSC following defibrillation may precipitate cardiac instability and re-arrest. May be avoided by using physiological monitoring such as quantitative waveform capnography, intra-arterial pressure monitoring, or continuous central venous oxygen saturation monitoring and avoiding administration of vasopressors if ROSC occurs (J Emerg Med 2009;38:614-21; Ann Emerg Med 1992;21:1094-101; Ann Emerg Med 1990;19:1104-6).

Patient Case

6. M.G., a 58-year-old woman with a history of chronic osteoarthritis and peptic ulcer disease, is admitted to the MICU with hypovolemic shock caused by a suspected bleeding gastric ulcer. Endoscopy is performed, confirming the gastric ulceration. The ulcer is cauterized, and M.G. is stabilized. On ICU day 2, M.G. becomes lethargic, hypoxic, and subsequently pulseless. The MICU team is summoned, and the monitor reveals VF. Resuscitation efforts begin, and the ACLS algorithm is followed. The patient will receive amiodarone. Which general principles are most accurate regarding amiodarone administration for M.G.'s VF arrest?
   A. Endotracheal delivery is preferred because all cardiac arrest medications can be delivered by the endotracheal route.
   B. Peripheral administration is preferred because the peak concentrations are higher and the circulation time is shorter than with other routes.
   C. Intraosseous administration is preferred because administration and dosing are similar to that for the endotracheal route.
   D. Central intravenous administration is preferred because the peak concentration and circulation time is shorter than with other routes.

c. Management of VF/pVT (Figure 1): Defibrillation (summary, details available previously in the IHCA and OHCA chain of survival: Defibrillation section)
   i. When VF/pVT detected, CPR should continue until defibrillator (either manual or automatic) charging period is over.
   ii. It is strongly recommended that CPR be performed while the defibrillator is readied because chest compressions can deliver oxygen and potentially unload the ventricles, increasing the likelihood that a perfusing rhythm will return after shock is delivered.
   iii. Because intentionally delaying defibrillation for CPR has mixed results, it cannot currently be recommended.
   iv. Once defibrillator is charged, patient is "cleared" (ensuring all members of the resuscitation team are no longer in contact with the patient).
   v. Shock is delivered, and CPR is immediately resumed, beginning with chest compressions.
      (a) Pulse check is delayed until 2 minutes of CPR is given.
      (b) Pause for rhythm and pulse check (less than 10 seconds); continue CPR, if necessary.
   vi. If biphasic defibrillator is available:
      (a) Provider should use manufacturer's recommended energy dose (e.g., 120–200 J).
      (b) If information unavailable, the maximum dosing can be considered.
      (c) Second and subsequent defibrillator energy dosages should be equivalent, and consideration should be made for escalating energy doses, if possible.

vii. If monophasic defibrillator is the only available defibrillator:
  (a) Initial energy dose should be 360 J.
  (b) Second and subsequent defibrillator energy dosages should be 360 J.

viii. If VF/pVT is terminated by defibrillation and reoccurs, resulting in an arrest, the successful energy dosage used previously should be considered.

ix. Change of multimodal defibrillator from automatic to manual mode may result in fewer interruptions of CPR but also an increased frequency of inappropriate shocks (Resuscitation 2007;73:131-6; Resuscitation 2007;73:212-20).

d. Medication therapy for VF/pVT

i. Previous guidelines suggested medication consideration after one shock and 2 minutes of CPR (one cycle). Data from witnessed OHCA suggest that early administration of epinephrine is associated with improved survival in shockable rhythms (Resuscitation 2015;96:180-5) and that every minute beyond 5 minutes significantly increases the odds of death. With IHCA for shockable rhythms, patients receiving epinephrine within 2 minutes of initial shock was associated with a decreased odds of survival, ROSC, and good functional outcome (BMJ 2016;353:i1577). Guideline recommendation supports the use of epinephrine for shockable rhythms after initial defibrillation attempts have failed (Circulation 2019;140:e881-e894).

ii. Vasopressors:
  (a) First-line medication is epinephrine. A randomized, double-blind, placebo-controlled trial showed that use of epinephrine in OHCA (around 20% shockable rhythms) was associated with greater 30-day survival, but there was no difference in neurologically favorable outcomes because of the severe neurologic impairment of survivors (N Engl J Med 2018;379:711-21).
    (1) A meta-analysis of randomized controlled trials found that epinephrine (1 mg every 3–5 minutes) when compared to placebo for OHCA increased rate of ROSC, survival to hospital admission and survival to hospital discharge but did not change discharge with favorable neurological function (Resuscitation 2019;139:106-21). The administration of epinephrine 1 mg every 3–5 minutes has been the standard medication within the cardiac arrest algorithm. An option to provide epinephrine every 4 minutes as a midrange has been added. This allows the provider to administer epinephrine every other 2-minute rhythm check (https://acls-algorithms.com/2021-aha-acls-guideline-changes/adult-cardiac-arrest-algorithm-changes/).
    (2) The traditional dose of epinephrine is considered 1 mg. Data analyses suggest that lower doses (e.g., 0.5 mg) are not associated with a change in survival to hospital discharge or favorable neurologic outcomes (Resuscitation 2018;124:43-8).
    (3) High-dose epinephrine (0.1–0.2 mg/kg) has been investigated and although it may increase the rate of ROSC, it has not been found to increase rate of survival to hospital admission, survival to hospital discharge, or survival with favorable neurological function (Circulation 2019;140:e881-e894).
  (b) Vasopressin has previously been removed from guidelines because of equivalence of effect with epinephrine and a drive to simplify treatment (Figure 1; Table 3). A meta-analysis affirms the equivalence of vasopressin with epinephrine (either in isolation or in combination), supporting the recommendation that vasopressin offers no advantage over epinephrine in cardiac arrest (Resuscitation 2019;139:106-21).
  (c) No other vasopressors (e.g., phenylephrine or norepinephrine) have shown any benefit compared with epinephrine (JAMA 1992;268:2667-72; Acta Anaesthesiol Scand 1985;29:610-3).

(d) Adding methylprednisolone and vasopressin to epinephrine during ACLS plus stress dose hydrocortisone for post-ROSC shock compared with placebo may aid in ROSC during cardiac arrest and improve discharge neurologic function in patients who survive. Role of steroids may be debated, given the addition of vasopressin in the study arm (JAMA 2013;310:270-9). A network meta-analysis suggests that the combination of vasopressin-steroids-epinephrine was the only vasopressor regimen associated with ROSC and, in IHCA, also survival (Crit Care Med 2018;46:e443-e451). Guidelines suggest this treatment bundle can be considered, but further confirmatory data are needed (Circulation 2015;132(suppl 2):S315-S367).

(e) Prehospital methylprednisolone in primarily shockable rhythms was studied in a phase II study and demonstrated significant reductions in inflammatory markers associated with post-cardiac arrest syndrome (Intensive Care Med 2024;49:1467-78). Further investigation may be warranted to determine the clinical impact of these findings.

iii. Antiarrhythmics:

(a) No evidence that, in cardiac arrest, any antiarrhythmics increase survival to discharge.

(b) Conflicting evidence on benefit of amiodarone and lidocaine (Table 3) in the prehospital arrest situation:

(1) Amiodarone or lidocaine may be considered for VF/pVT that is unresponsive to defibrillation (Circulation 2019;140:e881-e894).

(2) Investigation has found that amiodarone (300 mg) increased rates of survival to hospital admission compared to placebo in shock-refractory VF/pVT in those who received at least 3 shocks and epinephrine (N Engl J Med 1999;341:871-8). Additionally, 5 mg/kg amiodarone improved survival to hospital admission compared with 1.5 mg/kg lidocaine (N Engl J Med 2002;346:884-90). These 2 studies did not find evidence of improved discharge survival. Additionally, a comparison in patients with VF/pVT considered refractory after at least 1 shock, which found that ROSC was higher in patients receiving lidocaine compared with those receiving placebo, but not for those receiving amiodarone. Survival to hospital admission was higher with amiodarone or lidocaine compared to placebo, but no difference in survival with good neurological outcome or survival to hospital discharge in any group (N Engl J Med 2016;374:1711-22). These drugs may be particularly useful for patients with witnessed arrest, for whom time to drug administration may be shorter (Circulation 2019;140:e881-e894).

(3) Amiodarone should be administered as undiluted intravenous/intraosseous push if pulseless. Dosing should be 300 mg with repeat dosing of 150mg (Table 3).

(4) If pulse is obtained, the suggested dose is 150 mg, which must be given as slow intravenous piggyback over at least 15 minutes and is typically followed by a continuous intravenous infusion.

(5) Hemodynamic effects of bradycardia and hypotension may be partly related to vasoactive solvents (polysorbate 80) (Am J Cardiol 2002;90:853-9).

(c) Magnesium sulfate (Table 3) is effective for cardiac arrest caused by torsades de pointes. (i.e., caused by early afterdepolarizations during phase 2 of the action potential).

(1) Not effective when VF/pVT is not associated with torsades de pointes.

(2) May consider for emergency magnesium replacement in patients who sustain cardiac arrest and are hypomagnesemic.

(3) Optimal dosing has not been established.

(d) In patients with non-shockable rhythms that become shockable in OHCA, investigation has shown some numeric benefit of antiarrhythmics, perhaps warranting further investigation in this patient group (Circulation 2017;136:2119-31).

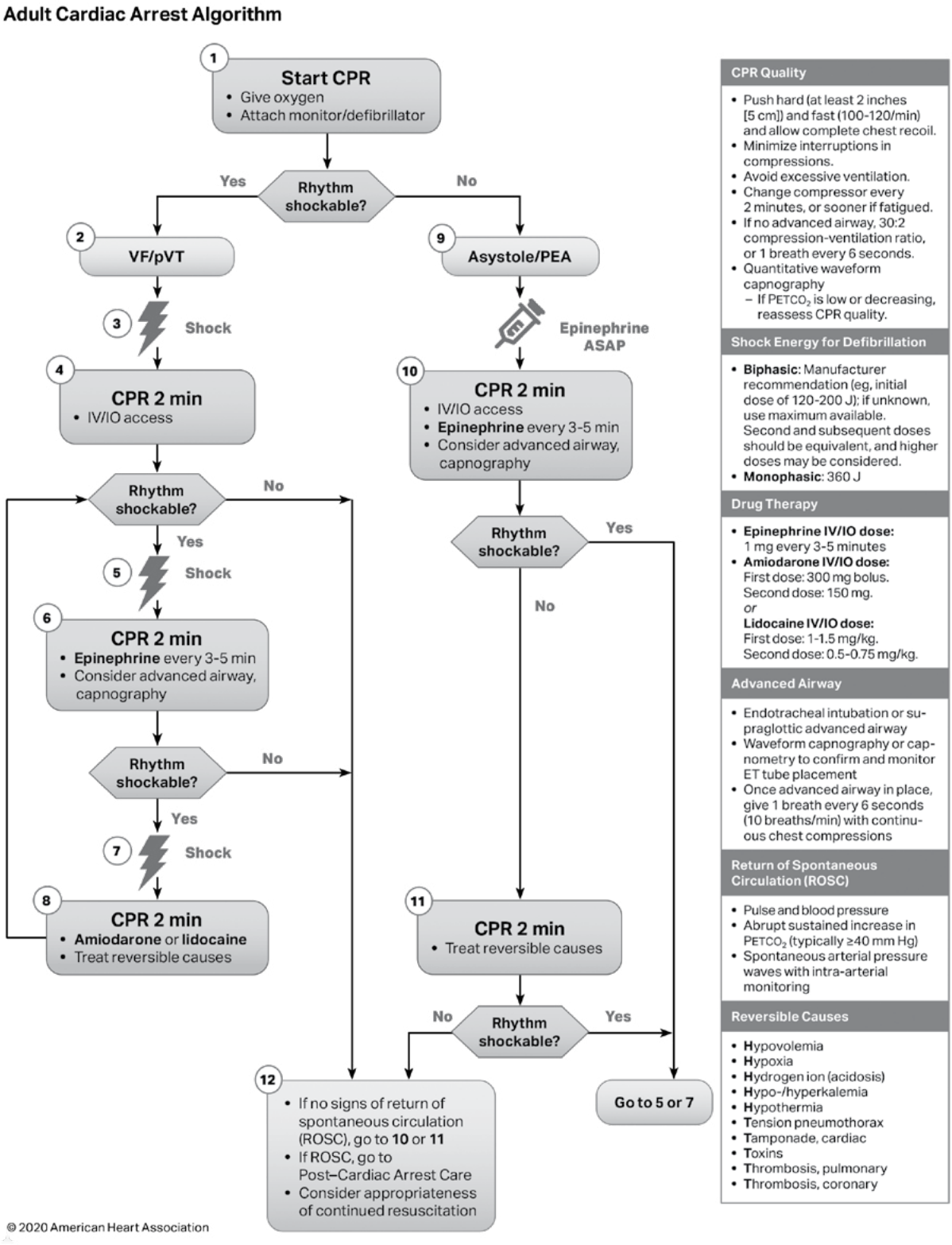


Figure 1. Adult cardiac arrest algorithm. CPR = cardiopulmonary resuscitation; IO = intraosseous(ly); IV = intravenous(ly); PEA = pulseless electrical activity; $PETCO_2$ = partial pressure of end-tidal CO2; pVT = pulseless ventricular tachycardia; VF = ventricular fibrillation.

Table 3. Medications Used During Sudden Cardiac Arrest

Medication	Primary Mechanism of Action in Cardiac Arrest	Dosage, Route, Frequency	Clinical Benefits
Epinephrine (N Engl J Med 1992;327:1045-50; Circulation 1984;69: 822-35; Crit Care Med 1979;7:293-6)	α-Adrenergic agonist effects leading to vasoconstriction	1 mg IV/IO q3–5 min 2–2.5 mg ET q3–5 min (diluted with either 5–10 mL of 0.9% sodium chloride or sterile water)	Increases coronary and cerebral perfusion pressure during CPR Increases ROSC Increases survival to hospital admission and discharge in OHCA
Amiodarone (N Engl J Med 2002;346:884-90; N Engl J Med 1999;341:871-8; N Engl J Med 2016;374:1711-22)	Na+/K+/Ca^2+ channel and β-receptor antagonist (Class III antiarrhythmic)	First dose: 300 mg or 5 mg/kg IV/IO x 1 Second dose: 150 mg IV/IO x 1 Max: 2.2 g/day	Conflicting data on clinical outcomes compared with lidocaine or placebo for OHCA
Lidocaine (Resuscitation 1997;33:199-205; N Engl J Med 2016;374:1711-22)	Na+ channel antagonist (Class IB antiarrhythmic)	First dose: 1–1.5 mg/kg IV/IO x 1 Subsequent dosing: 0.5–0.75 mg/kg q5–10 min Max 3-mg/kg cumulative dose	Conflicting data on clinical outcomes for OHCA; no improvement in overall or discharge survival Insufficient evidence to recommend for refractory VF/pVT unless amiodarone unavailable
Magnesium (Clin Cardiol 1993;16:768-74; New Trends Arrhythmias 1991;7:437-42; Circulation 1988;77:392-7)	Stops EAD in torsades de pointes by inhibiting Ca^2+ channel influx	1–2 g diluted in 10 mL of 5% dextrose or sterile water IV/IO × 1	Can aid in stopping torsades de pointes in patients with prolonged QT interval

EAD = early afterdepolarization; ET = endotracheal; IO = intraosseously; IV = intravenously; pVT = pulseless ventricular tachycardia; q = every.

e. Management of PEA/asystole
   i. CPR and treatment of reversible causes are vital to treatment of PEA/asystole.
   ii. Medication therapy
      (a) Epinephrine can be given as soon as feasible (Table 3).
         (1) Stepwise increased mortality is seen if first epinephrine dose withheld for more than 4 minutes after non-shockable IHCA (BMJ 2014;348:1-9). A randomized, double-blind, placebo-controlled trial found that use of epinephrine in OHCA (around 80% non-shockable rhythms) was associated with greater 30-day survival, but there was no difference in neurologically favorable outcomes because of the severe neurologic impairment of survivors (N Engl J Med 2018;379:711-21).
         (2) Recommended for all non-shockable rhythms to administer epinephrine as soon as feasible (Circulation 2015;132(suppl 2):S315-S367). Data analyses suggest that prompt (less than 5 minutes from non-shockable rhythms arrest) administration of epinephrine was associated with greater 1-year survival (Circulation 2018;137:2041-51). These results have been found at both the patient level and the hospital system level (Circulation 2016;134:2105-14). Recent guideline update supports the early usage of epinephrine as soon as feasible in the nonshockable rhythm context (Circulation 2019;140:e881-e894).
      (b) Atropine has been removed from the algorithm because of its lack of therapeutic benefit.

f. Role in treating reversible causes
   i. Echocardiography may be helpful, if available, in the management of PEA to help differentiate the following (Am J Cardiol 1992;70:1056-60):
      (a) Intravascular volume status (ventricular volume).
      (b) Cardiac tamponade.
      (c) Massive pulmonary embolism (right ventricular size, function).
      (d) Mass lesions (tumor, clot).
      (e) Coronary thrombosis (right and left ventricular function, regional wall motion abnormalities).
   ii. Because hypoxia is often a cause of PEA arrest, more focused attention may be given to placement of airway and oxygen delivery.
   iii. See the specific chapters for pulmonary disorders (massive pulmonary embolism and tension pneumothorax), cardiology (acute myocardial infarction and cardiac tamponade), shock (hypovolemic shock and oxygen delivery), acid-base disorders (acidemia), endocrinologic disorders (hypoglycemia), and electrolytes (hypo/hyperkalemia).

g. Controversial interventions in cardiac arrest
   i. Sodium bicarbonate
      (a) Tissue acidosis and acidemia result during cardiac arrest for several reasons, including inadequate or absent blood flow, arterial hypoxia, or underlying pathophysiology.
      (b) Mainstays of restoring acid-base status include high-quality chest compressions and appropriate ventilation/oxygenation.
      (c) Conflicting evidence exists for the use of sodium bicarbonate, with most data showing no benefit or poor outcome with use (Ann Emerg Med 1998;32:544-53; Resuscitation 1995;29:89-95; Am J Emerg Med 1992;10:4-7; Chest 1990;97:413-9; Resuscitation 1989;17(suppl):S161-172; discussion S199-206; J Emerg Med 2020;59:856-64).

(d) Data in patients with prolonged CPR efforts (> 20 mins) primarily from VF (~80%) demonstrated an association with increased rate of ROSC (Am J Emerg Med 2016; 34:225-9) after complex retrospective analysis. Caution should be used extrapolating results because significant limitations exist in the study design and all patients included likely were in the metabolic phase of VF (JAMA 2002;288:3035–8) and had pH values < 7.1. A prospective, observational, propensity-matched data suggest that OHCA use of prehospital sodium bicarbonate was associated with a decreased probability of favorable neurologic outcomes and survival (Resuscitation 2017;119;63-9).

(e) Detrimental effects may be associated with sodium bicarbonate in cardiac arrest, including:
  (1) Compromised coronary perfusion pressure by reducing systemic vascular resistance (JAMA 1991;266:2121-6).
  (2) Shifting the oxyhemoglobin dissociation curve to the left by creating an extracellular alkalosis and decreased release of oxygen.
  (3) Causing hypernatremia and subsequent hyperosmolarity leading to hyperviscosity, potentially impairing blood flow.
  (4) Producing excess CO_2 through rapid dissociation, which can freely diffuse intracellularly (e.g., myocardial and cerebral cells) and cause intracellular acidosis (Science 1985;227:754-6).
  (5) Inactivation of concurrently administered catecholamines (e.g., epinephrine) (Hosp Pharm 1969;4:14-22).

(f) Certain circumstances may warrant sodium bicarbonate use such as tricyclic antidepressant overdose, bicarbonate (HCO_3-)-wasting causes of metabolic acidosis, and hyperkalemia. In addition, it should be noted that endogenous catecholamines have a blunted response during acidosis which may improve if the acidosis is corrected. Initial dosage should usually be 1 mEq/kg intravenous push with monitoring of clinical status, HCO_3- concentration, laboratory values, and blood gas analysis.

(g) Recent meta analysis of RCT's found that use of sodium bicarbonate was not superior to control in regards to survival and ROSC and may be associated with lower rates of sustained ROSC and good neurological outcome (Int J Emerg Med 2021;14:21).

(h) A recent RCT in a porcine model of hyperkalemia-induced cardiac arrest discovered that sodium bicarbonate was associated with increased rates of ROSC and reduced time to ROSC (Crit Care Med 2024;52:e67-78). Human investigation may be warranted.

ii. Calcium

(a) No trial has established any impact on survival in either IHCA or OHCA (Ann Emerg Med 1985;14:626-9; Ann Emerg Med 1985;14:630-2). Recent RCT data in OHCA did not confer benefit of calcium on ROSC (JAMA 2021;326:2268-76).

(b) Consider in patients with preexisting hypocalcemia and signs and symptoms of acute hypocalcemia (e.g., severe tetany or seizures).

(c) A recent RCT in a porcine model of hyperkalemia-induced cardiac arrest discovered that calcium chloride was not associated with increased rates of ROSC or time to ROSC (Crit Care Med 2024;52:e67-78).

iii. Atropine

(a) No prospective studies have evaluated atropine for bradycardic PEA or asystolic cardiac arrest.

(b) Conflicting results exist from retrospective analyses and case reports (Acta Anaesthesiol Scand 2000;44:48-52; Ann Emerg Med 1984;13:815-7; Ann Emerg Med 1981;10:462-7).

(c) Atropine has not been associated with harm in treating bradycardic PEA or asystolic cardiac arrest, but because of the lack of convincing evidence of benefit, it is no longer recommended for cardiac arrest but is reserved for symptomatic/life-threatening bradycardia.

iv. Intravenous fluids
(a) Normothermic, hypertonic, and chilled fluids have been evaluated in animal models and small human studies, with no survival benefit.
(b) If hypovolemic shock is the suspected cause of the cardiac arrest, fluid resuscitation should be initiated immediately.
v. For indications of fibrinolysis for cardiac arrest, see the Pulmonary chapter for treatment of pulmonary embolism and the Cardiology chapter for treatment of acute myocardial infarction.
vi. Pacing
(a) Transcutaneous, transvenous, and transmyocardial pacing is not beneficial in cardiac arrest and does not improve ROSC or survival.
(b) Not recommended for routine use in cardiac arrest.
vii. Dextrose
(a) Animal data has demonstrated that dextrose administration before, during, or after cardiac arrest leads to higher rates of mortality and worse neurological outcomes (J Crit Care. 1987;2:4–14; Surgery 1990;72:1005–11; Acta Anaesthesiol 1986;100:505–11).
(b) Human retrospective data from IHCA demonstrated that dextrose administration during resuscitation was associated with a significantly decreased chance of survival to discharge and good neurological outcome (Crit Care 2015;19:160[1-8]).
viii. Cyclosporine - A randomized, single-blind trial found no impact on target organ damage, survival, or favorable neurologic function, contrary to experimental evidence (JAMA Cardiol 2016;1:557-65).

h. For information on acute symptomatic arrhythmias (bradycardias and tachycardias), see the Cardiovascular Critical Care I chapter.

D. Post–Cardiac Arrest Care and Recovery (IHCA and OHCA - chain of survival): Objectives of post–cardiac arrest care can be divided into initial (stabilization phase) and subsequent (continued management and additional emergent activities) (Circulation 2010;122:S768-786; Circulation 2008;118:2452-83; Circulation 2020;142:S366-S468).
1. Initial
a. Optimize hemodynamics: Target MAP 65–100 mm Hg, central venous pressure 8–12 mm Hg, central venous oxygen saturation greater than 70%, urine output greater than 1 mL/kg/hour, and normal serum lactate.
i. Consider the patient's normal BP, cause of arrest, and severity of myocardial dysfunction for all values above. Recent study has found that lower MAP goals (63 mm Hg vs. 77 mm Hg) did not portend worse survival or neurologic outcomes (N Engl J Med 2022;387:1456-66).
ii. Use intravenous crystalloids and colloids, continuous vasopressors and inotropes, transfusions, and renal replacement as needed to meet target goals.
iii. In a randomized study of vasopressor dependent shock post-cardiac arrest, the provision of corticosteroids (hydrocortisone 100 mg intravenously every 8 hours x 7 days or until shock reversal) did not improve time to shock reversal, rate of shock reversal, or clinical outcomes (Crit Care 2016;20:821-8]) and thus should be avoided unless otherwise indicated.
b. Transfer patient to a system (OHCA) or unit (IHCA) that can provide advanced post–cardiac arrest care, including continuous electrocardiographic (ECG) monitoring with immediate 12-lead ECG, central intravenous access if possible, coronary reperfusion, and/or temperature control (i.e., targeted temperature management).
c. Try to identify and treat the reversible causes of cardiac arrest (Table 2). Laboratory and diagnostic tests should be performed to aid in identifying a potential underlying cause.

2. Continued management and additional emergency activities
    a. Consider body temperature regulation. Should be considered for any patient with ROSC who does not follow commands (i.e., comatose) after cardiac arrest.
        i. Early evidence suggested significant benefit for primarily shockable (VF/pVT) OHCA (N Engl J Med 2002;346:549-56; N Engl J Med 2002;346:557-63; Circulation 2010;122:S768-786). A meta-analysis demonstrated a similar nonsignificant reduction in mortality and poor neurological outcome (Am J Med 2016;129:522-527) when including all randomized studies. After exclusion of one trial that allowed for temperature control in the control arm, there was a significant reduction in poor neurological outcome when implementing temperature control. A Cochrane analysis found that conventional temperature control compared with no temperature control conferred the benefit of favorable neurological outcome and survival with only a slight increase in the incidence of pneumonia and hypokalemia (Cochrane Database Syst Rev 2016;2:CD004128). The Targeted Temperature Management 2 (TTM2) study, published in 2021 comparing goal temperature of 33°C with normothermia (N Engl J Med 2021;384:2283-94), found similar outcomes surrounding neurologic function and overall survival. This has led some to advocate for normothermia (i.e., avoidance of fevers) versus hypothermia. In addition, in the TTM2 study, patients who underwent therapeutic hypothermia had more adverse events.
        ii. Evidence for nonshockable rhythms IHCA and OHCA has conferred results similar to those in early studies of shockable rhythms demonstrating increased survival and better neurological outcomes (N Engl J Med 2019;381:2327-37; Circulation 2015;132:2146-51).
        iii. Optimal targets, timing, duration, and other variables still unclear:
            (a) Goal target body temperature 32°C–37.5°C given data showing that targeting 36°C instead of 33°C may be equivocal (Circulation 2015;132(suppl 2):S315-S367; N Engl J Med 2013;369:2197-206; N Engl J Med 2002;346:549-56; N Engl J Med 2002;346:557-63; Circulation 2024;149:e254-73). Additional investigations have found that targeting a warmer temperature (36°C) may lead to reduced times in goal temperature range but with increased fever rates (Resuscitation 2017;113:39-43). Similar results have been found in other studies (Crit Care Med 2020;48:362-9; Resuscitation 2019;143:142-7). The TTM2 study, as mentioned earlier, has led some to consider strict avoidance of hyperthermia, using active maintenance of normothermia instead of targeting hypothermic temperature goals (N Engl J Med 2021;384:2283-94). European guidelines suggest a more modest application of literature and suggest avoiding fever while continuous monitoring core temperature (Intensive Care Med 2022;48:261-9).
            (b) Duration of at least 12 hours; optimally, a minimum of 24 hours. A 48-hour duration did not seem to confer any benefit compared with a 24-hour duration in a multicenter randomized study (JAMA 2017;318:341-50). Multiple studies are ongoing investigating the optimal duration of temperature control.
            (c) Initiate temperature control as soon as possible (within 2 hours, if possible), with goal temperature attainment within 6–8 hours; however, several retrospective studies have not confirmed the timing of initiation or the timing of temperature attainment as predictors of neurologic outcome (Acta Anaesthesiol Scand 2009;53:962-34; Int J Cardiol 2009; 133:223-8). Intra-arrest temperature control is currently being investigated, but human data is limited. Prehospital infusion of cold intravenous fluids for OHCA in several randomized controlled studies conferred no benefit and may increase the number of complications; thus, it is not recommended (Circulation 2015;132(suppl 2):S315-S367; Circulation 2016;134:797-805).
            (d) Modality for cooling includes feedback-controlled endovascular systems, surface-cooling devices, ice packs/bags, cooling blankets, and/or iced isotonic fluids.

(e) Axillary or oral temperature monitoring is inadequate for temperature control (Acta Anaesthesiol Scand 1998;42:1222-6; J Cardiothorac Vasc Anesth 1996;10:336-41), which requires central/core temperatures by esophageal, bladder (avoid in anuric patients), or pulmonary artery temperature monitoring. Ideally, the monitoring modality chosen will be used for other indications as well.

iv. The major complications of temperature control are related to the physiologic and adaptive responses to the temperature-changing process of temperature control (Table 4). The adverse events of sepsis, myoclonus, seizures, and hypoglycemia within 72 hours of temperature control have been associated with poor neurologic outcome (Crit Care 2015;19:283-96).

Table 4. Major Organ-Specific Complications of Therapeutic Hypothermia

Musculoskeletal (Crit Care Med 2015;43:2228-38; N Engl J Med 2013;369:2197-206; Anesthesiology 2009;111:110-5; Br J Anaesth 2005;94:756-62; N Engl J Med 2002;346:549-56; N Engl J Med 2002;346:557-63; JAMA 1997;277:1127-34; Am J Physiol 1960;198:481-6)	**Shivering:** Body's natural response to hypothermia, preceded by arteriovenous vasoconstriction; most common complication of hypothermia; can increase metabolic heat production by 600%, thereby slowing the induction of hypothermia. Typically slows or stops at core temperatures < 33.5°C 1. Agents that decrease the shivering threshold (i.e., decreases the temperature at which shivering will occur): a) Scheduled acetaminophen 650 mg q4–6 hr or buspirone 30 mg q12 hr reduces the shivering threshold by 0.2ºC–0.4ºC b) Magnesium sulfate reduces the shivering threshold by ~0.3°C c) Meperidine decreases the shivering threshold by up to 2°C but should be avoided because of decreased effective glomerular filtration rate (GFR) during hypothermia and increased risk of seizures when meperidine is used in patients with decreased GFR d) Dexmedetomidine and clonidine decrease the shivering threshold by ~0.8°C, but extreme caution should be used because of the hypotensive and bradycardic effects of both agents e) Propofol decreases the shivering threshold by ~0.6°C and has a linear relationship between serum concentrations and reduction in body temperature; caution should be used, given the hypotensive and bradycardic effects 2. Options to stop shivering include: a) Continuous or as-needed paralytics can be used for prevention and treatment of shivering (see the chapter on management of paralytics for appropriate selection and dosing of agent[s]) —Hypothermia decreases clearance and prolongs the duration of neuromuscular blockade (see Table 5 for examples) —Train of four is not a reliable method of monitoring during hypothermia; clinical monitoring or continuous EEG (electroencephalography) may be warranted b) Surface warming if using internal cooling devices c) Chilled fluids to promote faster core temperature reduction

Table 4. Major Organ-Specific Complications of Therapeutic Hypothermia *(continued)*

Neurologic (Circulation 2010;122:S729-767; Circulation 2008;118:2452-83; Pharmacotherapy 2008;28:102-11; Ther Hypothermia Temp Manag 2016;6:189-93)	**Sedation and Analgesia:** Adequate pain control and sedation must be used Target Richmond Agitation and Sedation Scale score of -3 to -5 (see chapter on sedation for agent selection and dosing) during hypothermia Accumulation of parent and active metabolites can be expected for each analgesic and sedative, which lead to prolonged sedation and potentially untoward adverse effects (see Table 5 for examples) **Seizures:** Possible complication of cardiac arrest and therapeutic hypothermia Consider benzodiazepines as first line to abort seizure Phenytoin, barbiturates, valproic acid, and propofol can all be used, with vigilant monitoring of adverse effects because of decreased clearance (see Table 5 for examples; for dosing and monitoring values, see the Status Epilepticus section in the Neurocritical Care chapter) **Intracranial Pressure:** Increases post-cardiac arrest likely because of ischemic injury and cytotoxic edema and is partially attenuated by temperature control Major increase appears to occur during rewarming phase of temperature control Uncertain if targeted therapy would impact outcomes
Cardiac (Heart Lung 2001;30:161-3; Int Anesthesiol Clin 1964;2:803-27)	**Arrhythmias:** Include VT, VF, and atrial fibrillation If life threatening, should consider discontinuing therapeutic hypothermia and active rewarming Sinus bradycardia is common during cooling and, in isolation, should not be treated unless it leads to hemodynamic instability (e.g., hypotension or organ dysfunction) **ECG observations:** Prolonged PR, QRS, and QT intervals Caution should be exercised when using medications that prolong the QT interval **Hemodynamics:** Decreased cardiac output Fluid shifts away from central compartment Vasoconstriction during cooling, which may lead to increased blood pressure Vasodilation during rewarming, which may lead to decreased blood pressure
Hepatobiliary (Pharmacotherapy 2008;28:102-11; Ther Drug Monit 2001;23:192-7; Anesthesiology 2000;92:84-93; Clin Pharmacol Ther 1979;25:1-7)	Elevated transaminases Reduced activity of non-cytochrome and cytochrome P450–mediated metabolism See Table 5 for examples of metabolic changes in selected medications during hypothermia

Table 4. Major Organ-Specific Complications of Therapeutic Hypothermia *(continued)*

Endocrinologic (N Engl J Med 2002;346:557-63; Endocrinology 1970;87:750-5)	Hyperglycemia caused by decreased insulin production and effect in periphery, increased gluconeogenesis, and glycogenolysis Continuous insulin infusions may be necessary for glucose control. Goal should be < 180 mg/dL without inducing hypoglycemia
Renal (Resuscitation 2004;60:253-61; J Neurosurg 2001;94:697-705)	Decreased effective glomerular filtration (urine output may increase because of cold diuresis, but not effective clearance) Electrolyte shifts (K+, PO_4^{3}-, Na+, Ca^{2}+) during cooling phase; reverse upon rewarming, thus caution should be exerted with replacement
Hematologic (Pharmacotherapy 2008;28:102-11; Br J Haematol 1999;104:64-8; Crit Care Med 1992;20:1402-5)	Coagulopathy caused by thrombocytopenia, impaired activation and activity of clotting factors, impaired platelet function Actively bleeding patients should not be cooled

Table 5. Pharmacokinetic and Pharmacodynamic Changes of Selected Medications During Hypothermia

Fentanyl	Plasma concentration increases by 25% with a 3.7-fold decrease in clearance
Morphine	Receptor affinity (μ) decreases as temperature decreases
Propofol	Plasma concentration increases by 28%; decreased clearance
Midazolam	Clearance decreases by about 11% per degree below 36.5°C
Rocuronium	Clearance decreases by 50%, increases duration of action 2-fold
Vecuronium	Clearance decreases by 11% per degree Celsius; increases duration of action 2-fold
Cisatracurium	Eliminated by Hofmann elimination, which is a temperature-dependent enzymatic process; anticipate prolonged activity
Phenytoin	AUC (area under the concentration-time curve) increases by 180%; clearance and elimination rate constant decrease by 50%

Information from: Pharmacotherapy 2008;28:102-11; Ther Drug Monit 2001;233:192-7; Anesthesiology 2000;92:84-93; Br J Haematol 1999;104:68-8; Eur J Anaesthesiol Suppl 1995;1:95-106; Clin Pharmacol Ther 1979;25:1-7)

v. Rewarming should be a passive process (around 0.33°C–0.5°C per hour) (Acta Anaesthesiol Scand 2009;53:926-34; N Engl J Med 2002;346:557-63; N Engl J Med 2002;346:549-56). Rapid rewarming can cause extreme electrolyte shifts and vasodilation with associated hypotension.

vi. Though the exact temperature goals and duration of temperature control is still being investigated, avoidance of fever and hyperthermia is clear and should be maintained during the first 48–72 hours after cardiac arrest (Arch Intern Med 2001;161:2007-12; Resuscitation 2013;84:1062-7; J Crit Care 2017;38:78-83; Intensive Care Med 2022;48:261-9).

b. Identify and treat acute coronary syndromes.

i. Cardiovascular disease and acute coronary ischemia are the most common causes of cardiac arrest (Am Heart J 2009;157:312-8; N Engl J Med 1997;336:1629-33).

ii. Consideration of treatment of acute coronary syndromes should not be deferred in patients who are comatose or when temperature control is used.

iii. Emergent coronary angiography is no longer recommended over a delayed/selective strategy in patients with ROSC after cardiac arrest. Exceptions exist for patients with ST-segment–elevation myocardial infarction, shock, electrical instability, signs of significant myocardial damage, or ongoing ischemia (Circulation 2024;149:e254-73). An analysis suggests that risk stratification and admission prognosis can help determine which patients will benefit from angiography (JACC Cardiovasc Interv 2018;11:249-56).

iv. Effects of antiplatelet and antithrombotic drugs augmented by derangements in normal platelet activation/coagulation pathways caused by hypothermia.

v. See the Cardiology chapter for the workup and treatment of acute coronary syndromes.

c. Optimize mechanical ventilation.

i. Goal arterial oxygen saturation is 94% or greater. A recent study found equivocal results with restrictive (Pao2 target of 68–75 mm Hg) versus liberal (Pao2 target of 98–105 mm Hg) oxygenation targets in comatose survivors of OHCA (N Engl J Med 2022;387:1456-66).

ii. Avoid hyperventilation or over-bagging to avoid increase in intrathoracic pressure and decrease in cardiac output.

iii. Goal Pa_{CO_2} is 40–45 mm Hg or P_{ETCO_2} 35–40 mm Hg.

d. Support organ systems

i. Vasopressor/inotropes to support end-organ perfusion (see Table 6)

(a) Adrenergic medications should not be administered with alkaline solutions or sodium bicarbonate because they are inactivated (Hosp Pharm 1969;4:14-22).

(b) Central line is recommended because any agent with α-agonist properties can lead to extravasation and tissue necrosis.

Table 6. Common Vasoactive Agents Used After Cardiac Arrest

Medication	Typical Dosing Range (mcg/kg/min)	Clinical Pearls
Epinephrine	0.03–0.3	Mixed α and $\beta_{(1>2)}$ activity Used to treat severe hypotension (e.g., SBP < 70 mm Hg) Used for symptomatic bradycardia Used for hemodynamically unstable anaphylactic reactions Higher doses associated with increased α_1 activity
Norepinephrine	0.03–0.3	$\alpha > \beta_{(1>2)}$ receptor activity Used to treat severe hypotension (e.g., SBP < 70 mm Hg) Should be used in volume-resuscitated patients First line for septic shock Higher doses associated with increased α_1 activity
Phenylephrine	0.3–3	Pure α-agonist Used to treat severe hypotension (e.g., SBP < 70 mm Hg) Should be used in volume-resuscitated patients Avoid in patients with low cardiac output

Table 6. Common Vasoactive Agents Used After Cardiac Arrest *(continued)*

Medication	Typical Dosing Range (mcg/kg/min)	Clinical Pearls
Dopamine	2–20	In general, dose-related receptor activity: 2–5 mcg/kg/min dopamine receptor, 5–10 mcg/kg/min β_1-receptor, > 10 mcg/kg/min α_1-receptor Does not provide exclusive receptor activity across dosing ranges and, thus, can be arrhythmogenic at any dose Use cautiously in patients with a history of heart disease or arrhythmias Useful for patients with bradycardia and hypotension
Dobutamine	2–20	Predominance of inotropic properties but with activity on $\beta_1 > \beta_2 > \alpha_1$-receptor Used to treat low cardiac output α_1-agonist and β_2-agonist counterbalance, leading to little change in systemic vascular resistance Can lead to vasodilation in select patients Less systemic or pulmonary vasodilation than milrinone More tachycardia than milrinone but similar risk of ventricular arrhythmias Use cautiously in patients with a history of arrhythmias
Milrinone	0.25–0.75	Phosphodiesterase type 3 inhibitor leading to increased intracellular cAMP leading to influx of calcium and subsequently inotropy and chronotropy Used to treat low cardiac output Loading dose rarely used because of significant systemic hypotension Longer duration of activity than dobutamine Accumulates in renal dysfunction More systemic and pulmonary vasodilation than dobutamine Less tachycardia than dobutamine but similar risk of ventricular arrhythmias Use cautiously in patients with a history of arrhythmias

[a]See chapter on shock for a detailed discussion regarding selection of agent, dosing, pharmacology, and clinical considerations.
SBP = systolic blood pressure

ii. Glucose management (Circulation 2010;122:S768-786; Circulation 2008;118:2452-83)
   (a) Avoidance of severe hypoglycemia (40 mg/dL or less)
   (b) Target moderate glucose control: 144–180 mg/dL
   (c) May require continuous insulin infusion to maintain goals

iii. Seizure control/prevention (Neurology 1988;38:401-5; JAMA 1985;253:1420-6): Myoclonus and seizures, or both, can occur in up to 35% of adult patients who achieve ROSC and are more common in those who remain comatose. Electrographic patterns seen in the patient population include electrographic seizures, status epilepticus, and ictal-interictal continuum patterns (Circulation 2024;149:e254-73).
   (a) Clonazepam, valproic acid, and levetiracetam are all effective for myoclonus, but clonazepam should be considered first line.
   (b) Benzodiazepines, phenytoin, valproic acid, propofol, and barbiturates are all effective for post–cardiac arrest seizures. Seizures and status epilepticus are treated similarly to non-cardiac arrest populations. For dosing and monitoring guidelines, see the Status Epilepticus section of the Neurocritical Care chapter.

(c) For patients with ictal-interictal continuum patterns on electroencephalography can be given nonsedating antiseizure medications (i.e., non-benzodiazepine).

iv. Renal dysfunction: The indications for initiating renal replacement therapy in cardiac arrest survivors are the same as in critically ill patients in general (Lancet 2005;365:417-30).

v. Infectious disease: Association has been shown between the use of prophylactic antibiotics and decreased incidence of pneumonia and sepsis (Resuscitation 2015;92:154-9) in patients who underwent temperature control to goal 32-34°C. Impact on length of stay or neurological outcome was not demonstrated and no information is available regarding impact of higher temperature goals on above outcomes.

e. Assess prognosis.

i. Brain injury and cardiovascular instability are the main determinants of survival after cardiac arrest (Intensive Care Med 2004;30:2126-8).

ii. If temperature control is used, a delay of 72 hours after rewarming should be implemented for prognostication.

iii. If temperature control is not considered:

(a) Prognostication should wait until 72 hours after ROSC.

(b) No clinical neurologic sign has shown to be predictive of neurologic prognosis less than 24 hours after arrest (Neurology 2006;66:62-8; Crit Care Med 1987;15:820-5).

iv. Prudent to perform any prognostication after removal of opioids, sedatives, paralytics, and so forth.

v. Using several modalities of testing including clinical examination, neurophysiological testing, and imaging is recommended.

f. Recovery – Care and support during recovery (Circulation 2020;142(16_suppl_2):S366-468)

i. Assist survivors with rehabilitation needs. Multimodal rehabilitation assessment and treatment of physical, neurologic, cardiopulmonary, and cognitive impairments are recommended before hospital discharge.

ii. Cardiac arrest survivors and their caregivers are recommended to receive comprehensive, multidisciplinary discharge planning, including medical and rehabilitative treatment recommendations, and to return to activity/work expectations.

iii. Structured assessment for anxiety, depression, posttraumatic stress, and fatigue is recommended for cardiac arrest survivors and their caregivers.

iv. New recommendations also call for provider debriefing and referral for emotional support, when needed, for those responding to and participating in resuscitation efforts.

Patient Case

7. K.G., a 71-year-old woman with a history of atrial fibrillation, coronary artery disease status post three-vessel coronary bypass artery grafting 6 years ago, diabetes, and osteoarthritis, is being admitted to your MICU for temperature control after PEA arrest and subsequent ROSC. K.G., who remained comatose after the ROSC, was intubated and is now hemodynamically stable (BP 94/72 mm Hg and HR 86 beats/minute). Which is the most accurate statement regarding temperature control?
   A. Hypoglycemia is a common complication of hypothermia and may require continuous dextrose infusions.
   B. Dose modifications or avoidance of cytochrome P450 (CYP)–metabolized medications should occur during hypothermia.
   C. The optimal duration of temperature control should be at least 72 hours to affect survival.
   D. Temperature targets should be 28°C–30°C to improve the neurologic recovery.

II. HYPERTENSIVE CRISIS

A. Definitions (Cardiol Clin 2012;30:533-43; Chest 2007;131:1949-62; Cardiol Clin 2006;24:135-46):
   1. Hypertensive urgency: A systolic blood pressure (SBP) of 180 mm Hg or greater and/or a diastolic blood pressure (DBP) of 110 mm Hg or greater without evidence of target organ damage.
   2. Hypertensive emergency: Presence of an abrupt significantly elevated BP (often defined as SBP greater than 200 mm Hg and/or DBP greater than 120 mm Hg) with concurrent target organ dysfunction (e.g., acute kidney injury, heart failure exacerbation, obtundation). Table 7 lists example conditions that, when accompanied by high BP, define hypertensive emergency.
   3. MAP: Average pressure in the arteries experienced during one cardiac cycle. Calculated by MAP = 1/3 SBP + 2/3 DBP.

Table 7. Examples of Acute Target Organ Damage and Clinical Presentations

Eclampsia, preeclampsia	Hypertensive encephalopathy
Acute kidney injury	Acute shortness of breath, flash pulmonary edema, or acute left ventricular dysfunction
Acute aortic dissection (type A or B)	Acute intracranial bleeding (nontraumatic)
Seizures	Acute myocardial ischemia/infarction
Retinopathy	Cerebral infarction

B. Estimated that 1-3% of patients with hypertension will experience a hypertensive crisis (Acta Med Scand Suppl 1981;650:1-62; Am J Cardiol 2011;108:1277-82),

C. 10-year survival approaches 70% (Q J Med 1993;96:485-93), with 1-year survival greater than 90%.

D. Common Causes (Cardiol Clin 2012;30:533-43; Cardiol Clin 2006;24:135-46):
   1. Intoxications – Cocaine, amphetamines, phencyclidine hydrochloride, stimulant diet pills
   2. Nonadherence to antihypertensive regimen
   3. Withdrawal syndromes (e.g., clonidine or β-antagonists)
   4. Drug-drug/drug-food interactions (e.g., monoamine oxidase inhibitors and tricyclic antidepressants, antihistamines, or tyramine)

5. Spinal cord disorders
6. Pheochromocytoma
7. Pregnancy

E. Management:
   1. Hypertensive urgency: Lower BP slowly during the first 24–48 hours using oral medications (often home regimen reinitiation). Does not require an ICU admission for treatment. Over-agressive treatment may occur if not appropriately differentiated from hypertensive emergency.
   2. Hypertensive emergency: Requires ICU monitoring and intravenous medications. See goals listed in Table 8.

Table 8. Time Interval for BP Lowering with Hypertensive Emergency[a]

Goal Time	BP Target
< 60 min	↓ DBP by 10%–15% or MAP by 25% with goal DBP ≥ 100 mm Hg
2–6 hr	SBP 160 mm Hg and/or DBP 100–110 mm Hg
6–24 hr	Keep above BP goals (hours 2–6) during first 24 hr
24–48 hr	Outpatient BP targets

[a]See compelling conditions in the text that follows.

   a. A 25% reduction in MAP during the first hour is targeted to maintain cerebral perfusion (blood flow autoregulation) and to not precipitate ischemia, which has been found with >50% reductions (Stroke 1984;15:413-6).
   b. If neurologic function deteriorates during the initial 25% decrease (or during subsequent lowering), therapy should be discontinued or a lesser percent decrease targeted (N Engl J Med 1990;323:1177-83).
   c. Compelling conditions leading to unique treatment timing and goals:
      i. Acute aortic dissection
         (a) Propagation of acute aortic dissection is dependent on arterial BP and shear stress (force of left ventricular contraction as a function of time).
         (b) HR and contractility control can minimize shear stress and, together with BP, become a target of management.
         (c) Goal HR less than 60 beats/minute and SBP less than 100 mm Hg as soon as possible (within 5–10 minutes).
      ii. Acute ischemic stroke
         (a) Hypertension with ischemic stroke is an adaptive response to maintain cerebral perfusion pressure to the brain.
         (b) Cerebral perfusion pressure (CPP) equals mean arterial pressure minus intracranial pressure (ICP): CPP = MAP – ICP.
         (c) Treatment should occur only if (1) thrombolytic therapy is required (goal SBP less than 185 mm Hg and DBP less than 110 mm Hg before thrombolysis initiation - has been shown to decrease risk of bleeding), (2) other target organ damage occurs, or (3) SBP is greater than 220 mm Hg and/or DBP is greater than 120 mm Hg (Stroke 2013; 44: 870-947).
         (d) If treatment in indicated (outside of thrombolytic goal stated above): goal 10%–20% MAP reduction over 24 hours (Cardiol Clin 2012;30:533-43; CNS Drugs 2015; 29: 17-28).

iii. Intracerebral hemorrhage
(a) BP reduction goals will be based on individual factors, including medical history; ICP, if known; demographics such as age; presumed cause of hemorrhage (e.g., arteriovenous malformation); and interval since onset.
(b) High BP is associated with worse outcomes, including hematoma expansion, neurological deterioration, death, and inability to perform activities of daily living after intracranial hemorrhage (Eur J Neurol 2013;20:1277-83; Stroke 2013;44:1846-51; J Hypertens 2008;26:1446-52).
(c) BP reductions in patients without ICP elevations to goal SBP targets of less than 140 mm Hg and/or less than 160 mm Hg have been shown to be safe and may confer benefit regarding functional recovery (Stroke 2013;44:1846-51; Stroke 2012;43:2236-8; Arch Neurol 2010;67:570-6; Hypertension 2010;56:852-8; Lancet Neurol 2008;7:391-9; N Engl J Med 2013; 368: 2355-2365). In a study evaluating an early (within 6 hours) intensive care bundle that included the early intensive lowering of systolic blood pressure (target less than 140 mm Hg), strict glucose control in those without diabetes, antipyrexia treatment (target body temperature less than or equal to 37.5°C), and rapid reversal of warfarin-related anticoagulation (target international normalized ratio less than 1.5) within 1 hour, patients had improved functional outcomes and fever adverse events (Lancet 2023;402:27-40). Although promising, given the bundled nature of the intervention, it is hard to know what portion of the care bundle was driving the benefits seen.
(d) It is unclear whether aggressive targets are safe in patients with extreme elevations in BP (i.e., SBP greater than 220 mm Hg), patients with large hematomas, or those with elevations in ICP. Aggressive therapy can be considered, though a more modest reduction to an SBP less than 180 mm Hg or a MAP less than 130 mm Hg over the first 24 hours is recommended (Stroke 2015;46:2032-60).

3. Agents for BP management of hypertensive emergency
a. The drugs of choice for many presentations of hypertensive emergency are intravenous nicardipine or nitroprusside (Neurology 2014;83:1523-9; Curr Hypertens Rep 2014;16:450; High Blood Press Cardiovasc Prev 2022;29:33; Hypertension 2020;75:1334-57).
i. Intravenous nitroprusside works rapidly and is safe in the presence of renal and/or hepatic impairment for short-term use (24 hours or less).
ii. Continuous BP monitoring (e.g., arterial line) is recommended with use because rapid changes can occur.
iii. Nitroprusside can increase ICP and may result in coronary steal; caution or avoidance should be considered in patients with elevated ICP and acute myocardial ischemia/infarction.
b. Table 9 summarizes available agents, dosing, onset, duration, and hemodynamic considerations. Table 10 summarizes possible indications and special considerations.

Table 9. Medications Used in Hypertensive Emergencies

Medication	Usual Dosing Range	Onset	Duration	Preload	Afterload	Cardiac Output
Nitroprusside	IV 0.25–10 mcg/kg/min Titrate by 0.1–0.2 mcg/kg/min q5 min	Seconds	1–2 min	↓	↓↓	↑
Hydralazine	IV bolus: 10–20 mg IM: 10–40 mg q30 min PRN	IV: 10 min IM: 20 min	IV: 1–4 hr IM: 2–6 hr	↔	↓	↑
Nicardipine	IV 5–15 mg/hr Titrate by 2.5 mg/hr q5–10 min	5–10 min	2–6 hr	↔	↓	↑
Clevidipine	IV 1–6 mg/hr Titrate by 1–2 mg/hr q90s or consider doubling the dose every 90 seconds.. Max 32 mg/hr	1–4 min	5–15 min	↔	↓	↑
Nitroglycerin	IV 5–200 mcg/min Titrate by 5–25 mcg/min q5–10 min	2–5 min	5–10 min	↓↓	↓↔	↔↑
Esmolol	IV 25–300 mcg/kg/min (bolus of 500 mcg/kg, not often required given short onset) Titrate by 25 mcg/kg/min q3–5 min	1–2 min	10–20 min	↔	↔	↓
Metoprolol	IV bolus: 5–15 mg q5–15 min PRN	5–20 min	2–6 hr	↔	↔	↓
Labetalol	IV bolus: 20 mg, may repeat escalating doses of 20–80 mg q5–10 min PRN IV 1–2 mg/min Titrate by 1–2 mg/min q2 hr given longer half-life	2–5 min, peak 5–15 min	2–6 hr	↔	↓	↓
Enalaprilat	IV bolus: 1.25 mg q6 hr Titrate no more than q12–24 hr; max dose 5 mg q6 hr	15–30 min	12–24 hr	↓	↓	↑
Phentolamine	IV bolus: 1–5 mg PRN; max 15 mg	Seconds	15 min	↔	↓	↑
Fenoldopam	IV 0.03–1.6 mcg/kg/min Titrate by 0.05–1 mcg/kg/min q15 min	10–15 min	10–15 min	↔↓	↓	↑

IM = intramuscular; IV = intravenous; PRN = as needed.

Table 10. Indications and Special Considerations for Medications Used for Hypertensive Emergencies *(continued)*

Medication	Indication	Special Consideration
Nitroprusside	Most indications (exclusions: ICP elevation, coronary infarction/ischemia)	Liver failure – cyanide accumulation Renal failure – thiocyanate accumulation Can draw serum cyanide and thiocyanate concentrations to monitor Toxicity associated with prolonged infusions (> 72 hr) or high doses (> 3 mcg/kg/min) May result in coronary steal Increases ICP
Hydralazine	Pregnancy	Can result in prolonged hypotension (less predictable dose response) Risk of reflex tachycardia Headaches, lupus-like syndrome (with long-term use)
Nicardipine	Most indications, especially acute ischemic or hemorrhagic stroke	Risk of reflex tachycardia Infusion can lead to large fluid volumes administered
Clevidipine	Most indications, especially acute ischemic or hemorrhagic stroke	Formulated in oil-in-water formulation providing 2 kcal/mL of lipid calories Caution for patients allergic to soy or eggs
Nitroglycerin	Coronary ischemia/ infarction Acute left ventricular failure Pulmonary edema	Tachyphylaxis occurs rapidly, requiring dose titrations Adverse effects: Flushing, headache, erythema; often dose-limiting adverse effects Veno > arterial vasodilator
Esmolol	Aortic dissection Coronary ischemia/ infarction	Contraindicated in acute decompensated heart failure Should be used in conjunction with an arterial vasodilator for BP management in aortic dissection (initiate esmolol first because of the delayed onset relative to vasodilators such as nitroprusside) Metabolism is organ-independent (hydrolyzed by esterases in blood) Useful in tachyarrhythmias
Metoprolol	Aortic dissection Coronary ischemia/ infarction	Contraindicated in acute decompensated heart failure Should be used in conjunction with an arterial vasodilator for BP management in aortic dissection (initiate metoprolol first because of the delayed onset relative to vasodilators such as nitroprusside) Useful in tachyarrhythmias
Labetalol	Aortic dissection Coronary ischemia/ infarction Pregnancy	May be used as monotherapy in acute aortic dissection Contraindicated in acute decompensated heart failure Prolonged hypotension and/or bradycardia may be experienced with over-treatment; dose cautiously
Enalaprilat	Acute left ventricular failure	Contraindicated in pregnancy Caution in dose adjustments given prolonged duration of action

Table 10.*(continued)*

Medication	Indication	Special Consideration
Phentolamine	Catecholamine excess (e.g., pheochromocytoma)	Use in catecholamine-induced hypertensive emergency
Fenoldopam	Most indications	Risk of reflex tachycardia Caution with glaucoma Can cause hypokalemia, flushing May increase ICP

ICP = intracranial pressure.

c. Certain populations require specific medication therapy approaches:
   i. Pregnancy
      (a) Definitions of hypertensive disorders in pregnancy (Obstet Gynecol 2013;122:1122-31; Obstet Gynecol 2019;133:e26-50; Obstet Gynecol 2020;135:1492-5):
         (1) Hypertension in pregnancy: SBP greater than or equal to 140 mm Hg or DBP greater than or equal to 90 mm Hg, or both, measured on two occasions at least 4 hours apart
         (2) Gestational hypertension: SBP greater than or equal to 140 mm Hg or DBP greater than or equal to 90 mm Hg, or both, measured on two occasions at least 4 hours apart diagnosed after 20 weeks of gestation and a previously normal BP
         (3) Severe-range hypertension: SBP greater than or equal to 160 mm Hg or DBP greater than or equal to 110 mm Hg, or both, measured on two occasions at least 4 hours apart
         (4) Preeclampsia: Gestational hypertension or severe-range hypertension, in addition to at least 1 of the following:
            (A) Proteinuria (greater than or equal to 300 mg/24-hour urine, or protein:creatinine greater than or equal to 0.3, or dipstick 2+ only if other quantitative methods not available)
            (B) Renal insufficiency (serum creatinine greater than 1.1 mg/dL or doubling of the serum creatinine concentration in the absence of other renal disease)
            (C) Thrombocytopenia (greater than 100 ×109/L)
            (D) Impaired liver function (ALT/AST greater than or equal to two times upper limit of normal)
            (E) Pulmonary edema
            (F) New-onset headache or visual disturbances (not caused by alternative diagnoses)
         (5) Preeclampsia with severe features: Severe-range hypertension and:
            (A) Renal insufficiency
            (B) Thrombocytopenia
            (C) Impaired liver function
            (D) Severe persistent right upper quadrant or epigastric pain unresponsive to medications
            (E) Pulmonary edema
            (F) New-onset headache or visual disturbances
         (6) Eclampsia: new-onset tonic-clonic, focal, or multifocal seizures in the absence of other causative conditions often associated (but not required) with hypertension in pregnancy
      (b) Preeclampsia with severe features can only be managed by delivery of the baby.
      (c) Magnesium can be considered as an adjunctive therapy to decrease seizure risk or if seizures develop.

(d) Intravenous medications should only be considered for (1) severe-range hypertension, (2) preeclampsia/eclampsia, and/or (3) hypertensive emergency (signs of target organ damage) in a pregnant patient.
(e) Hydralazine and labetalol are feasible first-line options, and labetalol may have fewer adverse effects (Am J Health Syst Pharm 2009;66:337-44). Choice of agent should be dependent on provider familarity and local practices and protocols (Cochrane Database Syst Rev 2013;2013:CD001449).
(f) BP target goal for severely elevated BPs, preeclampsia/eclampsia, and hypertensive emergency in a pregnant patient is 160/110 mm Hg or less with avoidance of abrupt decreases in BP, which can lead to potentially harmful fetal effects. Because of this caution, the MAP should be decreased by ~20%-25% over the first few minutes to hours, and the BP should then be further decreased to the target of 160/110 mm Hg or less over the subsequent hours.

ii. Catecholamine-induced hypertensive emergency
(a) Phentolamine is the drug of choice because it competitively inhibits α-adrenergic receptors.
(b) β-Selective antagonists are contraindicated unless the patient is fully α-blocked.

iii. Cocaine-induced hypertensive emergency (Ann Emerg Med 2008;52:S18-20; Chest 2007;131:1949-62)
(a) Benzodiazepines are used to target the central effects of cocaine as first-line therapy and often will result in control of tachycardia and hypertension. Can consider diazepam 5–10 mg intravenously or lorazepam 2–4 mg intravenously titrated to effect.
(b) If central control of cocaine-induced hypertension fails, consider direct α-antagonism with phentolamine.
(1) Phentolamine 1 mg intravenously; repeat every 5 minutes as needed.
(2) If direct α-antagonism does not gain control or is unavailable, consider additional antihypertensives:
(A) Nitroglycerin, nicardipine, nitroprusside, or fenoldopam titrated to effect are viable options (see Tables 9 and 10 for dosing and considerations).
(B) Verapamil and diltiazem decrease coronary vasospasm associated with acute cocaine intoxication (Am J Cardiol 1994;73:510-3).
(C) Controversy exists regarding the use of β-blockers.
- Labetalol has shown conflicting results regarding ability to control MAP but does not alleviate cocaine-induced coronary vasoconstriction.
- Consensus opinion recommends β-selective antagonists only if full α-antagonism is used first.

iv. Blood pressure variability (BPV)
(a) Concept of BPV emerging as a key therapeutic target in various populations (Stroke 2014;45:2275-9; Eur J Neurol 2013;20:1277-83; J Cardiothorac Vasc Anesth 2014;28:579-85).
(b) BPV is often expressed as the standard deviation of SBP, MAP, DBP, or the area under the curve of time spent outside of blood pressure target.
(c) In the intracerebral hemorrhage population, decreased BPV has been correlated with improved early neurological function (Eur J Neurol 2013;20:1277-83), favorable outcome (Stroke 2014;45:2275-9), and decreased death or major disability (Lancet Neurol 2014;13:364–73).
(d) Agent selection can influence BPV.
(1) Compared with labetalol, nicardipine has demonstrated superior results regarding BPV (Neurocrit Care 2013;19:41-47; Neurocrit Care 2008;9:167-176), but impact on clinical outcomes has not been demonstrated.

(2) In the cardiac surgery population, clevidipine demonstrated better perioperative BPV compared with nitroglycerin or sodium nitroprusside but not compared with nicardipine (Anesth Analg 2008;107:1110-1121) and was associated with decreased time to extubation and postoperative length-of-stay (J Cardiothorac Vasc Anesth 2014;28:579-85).

(e) Unclear what the future clinical impact will be on other populations, how agent selection will influence BPV, but should be an area of future research.

d. All intravenous medications should be transitioned to oral medications as soon as possible.

i. Oral antihypertensives should be initiated within 24 hours.

ii. Medication history and reconciliation can assist in resuming home regimens.

iii. Additional or new agents should be selected according to disease-specific indications.

Patient Case

8. B.B. is a 44-year-old man with no significant medical history who presents to the ED with a ripping sensation in his chest. His social history includes cigarette smoking, 1.5 packs/day for the past 20 years. Chest radiography in the ED reveals mediastinal widening. Cardiac enzymes are within normal limits. Laboratory values include sodium 135 mEq/L, potassium 4.3 mEq/L, HCO_3- 24 mEq/L, SCr 0.55 mg/dL, glucose 110 mg/dL, DBil 0.2 mg/dL, and AST 39 U/L. B.B. is rushed for a chest CT and angiography, which reveal an acute type A and B aortic dissection. His vital signs include BP 208/140 mm Hg and HR 120 beats/minute. Which is the most appropriate management for B.B.?

A. Use esmolol to achieve goal of 25% reduction in MAP during the first 60 minutes.

B. Use lisinopril and hydrochlorothiazide to achieve BP reduction goal to 160/100 mm Hg during the first 24 hours.

C. Use labetalol to achieve 25% reduction in MAP and HR less than 60 beats/minute in the first 60 minutes.

D. Use esmolol with or without nitroprusside to achieve SBP less than 100 mm Hg and HR less than 60 beats/minute as soon as possible.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**

Because rescuer fatigue is common and may lead to inadequate compression quality, it is recommended to change rescuers every 2 minutes, with no more than 5 seconds between changes (Answer A is incorrect). Compressions are vital because they increase intrathoracic pressure and directly compress the heart, leading to oxygen delivery to the vital organs (Answer B is correct). Specific aspects of chest compression quality are necessary. These include a rate of 100–120 compressions per minute at a depth of 2–2.4 inches in adults, allowing for recoil after each compression; placement of the patient on a hard surface (e.g., backboard); and minimization of interruptions (Answer C is incorrect). Outcomes, including neurologically intact survival, ROSC, and possibly overall survival, are linked to minimizing interruptions in chest compressions. Because of this, it is recommended that interruptions (e.g., pulse checks and intubation) be less than 10 seconds and that chest compressions be resumed immediately (Answer D is incorrect).

2. **Answer: C**

Cardiac arrest patients have minimal blood flow, and oxygenation/ventilation requirements are lower; the new recommendation is to provide 1 breath every 5–6 seconds (10–12 breaths/minute) until an advanced airway is in place (Answer C is correct). Although the optimal ratio is unclear and chest compressions appear to be more vital to resuscitation, other ratios cannot currently be recommended (Answer A is incorrect). It is clear, however, that excessive ventilation can lead to decreased venous return and gastric inflation, which can lead to aspiration, regurgitation, and impacts on outcomes. In this case, a bag-mask ventilator is available, and more than one rescuers are involved, so the bag-mask ventilator should be used (Answer B is incorrect). In single health care provider rescuer situations, the bag-mask ventilator should never be used, and mouth-to-mouth or mouth-to-barrier resuscitation is recommended. In a single nonmedical rescuer situation, hands-only CPR is recommended. Advanced airways can be considered but should be placed only by experienced and trained personnel. Bag-mask ventilation can provide adequate oxygenation/ventilation until an airway can be secured (Answer D is incorrect).

3. **Answer: A**

Three vital actions with VF aid in survival: call emergency response team (already accomplished in case), begin CPR (must be initiated in case), and deliver shock (must occur in case) (Answer A is correct). Pacing can be effective in overriding stable VT but should not be used in the cardiac arrest or hemodynamically unstable patient (Answer B is incorrect). It is currently unclear whether postponing defibrillation for the provision of chest compressions first is of benefit, but it is clear that chest compressions should be initiated until the defibrillator is ready, charged, and set to deliver the shock because this increases the likelihood of success with defibrillation (Answer C is incorrect). Because time in VF predicts survival, and the longer patients are in VF the more difficult it is to terminate the arrhythmia, alternative treatments such as medications should not impede the provision of defibrillation (Answer D is incorrect).

4. **Answer: D**

After the advanced airway is in place, it is crucial to confirm placement in order to provide the intended oxygenation/ventilation. The confirmation should occur with both clinical and objective measurements (Answer D is correct). These include a physical assessment of the chest and epigastrium, end-tidal CO_2 monitoring, and/or continuous waveform capnography. In most cardiac arrests (particularly in this patient's pVT), airway management should not impede the provision of CPR and/or defibrillation (when defibrillation is indicated) (Answer B is incorrect). After the advanced airway is in place, 100% oxygen should be delivered to optimize the arterial oxygen saturation (Answer C is incorrect). In the cardiac arrest population, this has not been shown to carry the same toxicity as in other populations. Furthermore, after advanced airway is placed, compressions should be administered at a rate of 100–120 compressions per minute continuously, with breaths every 6 seconds (Answer A is incorrect).

5. **Answer: B**

In all cardiac arrests, the treatable causes (i.e., H's and T's) should be reviewed and addressed, if possible (Answer B is correct). In patients for whom the laboratory and diagnostic data are known, the information should be reviewed while CPR is being provided. In

patients for whom the information is unknown, clinical evaluation and attainment of information should occur, when possible. This retrieval of information, together with the administration of medications and advanced airway placement, should never impede on the provision of CPR or defibrillation, if indicated, because defibrillation (for VF and pVT)) and CPR are the only strategies that have been shown to affect survival from cardiac arrest (Answer A is incorrect). Pulseless electrical activity is not a wide complex dysrhythmia, and defibrillation would not be indicated (Answer C is incorrect). Post-cardiac arrest care is crucial in the prevention of re-arrest and, in temperature control, can significantly affect neurologic outcomes (Answer D is incorrect).

6. **Answer: D**

In general, it is important to remember that medication administration benefits only myocardial blood flow, ROSC, and possibly survival to hospital admission in cardiac arrest. Medication administration should never impede the provision of CPR and/or defibrillation. Central administration, if already available, is preferred for several reasons. These include higher peak concentrations, shorter circulation time, more standard dosing, and the lack of additional administration techniques needed (Answer D is correct). Endotracheal administration is an option, but only NAVEL (naloxone, atropine, vasopressin, epinephrine, and lidocaine) medications can be administered, the optimal doses are unknown, and medications must be diluted before administered (Answer A is incorrect). Given that this patient is in VF arrest, amiodarone, for example, could not be administered by endotracheal administration if it were indicated. Peripheral administration can be used and is the most common route of administration given that most hospitalized patients have it already, but it requires an additional bolus of fluid afterward and has a longer circulating time than does central administration (Answer B is incorrect). Intraosseous can also be used, with the caveat that tibial intraosseous administration is similar to peripheral administration and requires training to master the technique for placement (Answer C is incorrect).

7. **Answer: B**

Temperature control improves neurologic recovery when initiated, optimally within 2 hours but in up to 6–8 hours after VF cardiac arrest (application to all forms), and used for 12–24 hours (Answer C is incorrect). The goal temperature is 32°C–36°C. Data suggest there is no benefit of 33°C versus 36°C in improvement of survival or neurologic outcomes (Answer D is incorrect). If temperature control will be used, close monitoring of complications should occur. These complications include hyperglycemia caused by decreased insulin production and peripheral activity (Answer A is incorrect), bradycardias, enzymatic slowing (including CYP system), increased incidence of sepsis and infections, coagulopathies, decreased glomerular filtration, and shivering (Answer B is correct).

8. **Answer: D**

This patient is experiencing a hypertensive emergency with his target organ damage being an acute aortic dissection. Aortic dissection is one of the individual hypertensive emergencies that has a specific mechanism of worsening (propagation) from BP and shear stress, which require both rapid BP and HR control. Given the gravity of propagation, goals for aortic dissection are HR less than 60 beats/minute and SBP less than 100 mm Hg within minutes, if possible (Answer A is incorrect). This can be accomplished with a single agent like labetalol, which will control HR with its β-antagonist properties and decrease BP (afterload) with its α-antagonist properties (Answers B and C are incorrect). Esmolol can also be used as first line but will likely require an additional afterload-reducing agent such as nitroprusside (Answer D is correct).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**
The largest portion of adult cardiac arrests are caused by cardiovascular, not respiratory, events (Answer A is incorrect). In addition, this patient was on room air and nasal cannula immediately before the event, which could suggest his respiratory status was stable and unlikely to lead to cardiac arrest. Advanced airways and medications have only been shown to facilitate ROSC in cardiac arrest in contrast to chest compressions and defibrillation (if indicated), which can improve survival. Because of this, CPR should begin immediately for this patient before line placement to facilitate medication delivery (Answer B is incorrect), starting with chest compressions in accordance with the BLS guidelines with pads placed simultaneously to facilitate rapid defibrillation if the patient's rhythm reveals a shockable rhythm (Answer C is correct; Answer D is incorrect).

2. **Answer: A**
The cornerstone of therapy for VF cardiac arrest is rapid defibrillation (Answer A is correct). The recommended dosage of voltage for biphasic defibrillators is 200 J or the manufacturer's recommendation (often the same dosage). Although amiodarone is in the treatment algorithm for VF cardiac arrest, it is reserved and recommended for refractory VF cardiac arrest, which is defined as defibrillation refractory (Answer B is incorrect). Therefore, defibrillation should occur first. Atropine has been removed from the cardiac arrest algorithms for PEA and asystole because of a lack of benefit on outcomes and should not be considered for VF cardiac arrest (Answer C is incorrect). Pacing has not shown benefit in cardiac arrest and should not be used (Answer D is incorrect).

3. **Answer: B**
Because the rhythm detected is PEA and no longer VF, the cornerstone of therapy changes from giving defibrillation (Answer A is incorrect) to giving high-quality chest compressions and addressing the treatable causes of cardiac arrest (H's and T's) (Answer B is correct). Lidocaine is reserved for refractory VF/pVT) (defined as defibrillator refractory) when amiodarone is unavailable (Answer C is incorrect). Atropine was previously recommended for PEA/asystole, but in the 2010 ACLS guidelines it was removed because of a lack of data supporting any beneficial outcomes (Answer D is incorrect).

4. **Answer: C**
Temperature control is a consideration, according to the international guidelines for all patients with ROSC who remain comatose after a cardiac arrest (Answer A is incorrect). Although most well-designed and executed studies primarily enrolled patients with VF cardiac arrest, application is recommended for all patients with a cardiac arrest independent of rhythm. Although worsening transaminitis and hepatic enzymatic function slowing is likely to occur during hypothermia, neither of these principles would be considered a contraindication for hypothermia (Answer B is incorrect). Furthermore, renal function with respect to glomerular filtration worsens, which requires vigilant monitoring of renal function and close attention to renal dose modifications and serum concentration monitoring of medications when applicable (Answer C is correct). An additional complication of therapeutic hypothermia is an induced coagulopathy, which leads patients to be at risk of bleeding. Thrombolytics may carry specific indications for cardiac arrest (e.g., pulmonary embolism or acute coronary syndromes), but they would not be recommended empirically (Answer D is incorrect).

5. **Answer: B**
Temperature control improves neurologic recovery in patients after a cardiac arrest (Answer B is correct). Most patients included in randomized clinical trials have VF as the causative rhythm (Answers A and D are incorrect). Because of its impact on neurologic recovery, guideline recommendations have applied this literature to cardiac arrests of all rhythms, and many institutions have adopted this same recommendation. Recent studies targeting mild hypothermia (36°C vs. 33°C) have shown no difference regarding outcomes between the two modalities, and because of this, some question the utility of hypothermia at all in an era of potentially more advanced cardiac arrest care (Answer C is incorrect).

6. **Answer: C**
By definition, this patient is having a hypertensive emergency because she has an abrupt, severe increase in BP with target organ damage—in this case, potentially shock liver and vision changes. (Answer A is incorrect). The patient's SCr is not elevated from baseline and is thus not presently showing target organ damage (Answer

B is incorrect). This patient's D-dimer is normal as well, but even if it were elevated, this is not a specific marker of target organ damage (Answer D is incorrect). The patient has significant elevations in her transaminases, indicating injury to the liver and thus qualifying her for hypertensive emergency (Answer C is correct).

7. **Answer: C**

The patient should be initiated on nitroprusside for BP management (Answer C is correct). Although she has transaminitis, nitroprusside can be used safely in the first 24 hours in these patients. Cyanide accumulation would be a concern with extended use, but for the first 24 hours, the patient will be at minimal risk, and the benefit of rapid BP control outweighs any potential risk. Phentolamine would be reserved for hypertensive crisis that presents from a catecholamine crisis (Answer A is incorrect). Oral metoprolol might be an appropriate option to transition to after the emergency is resolved, but because of the target organ damage, more rapid reduction is needed using an intravenous agent such as nitroprusside (Answer B is incorrect). Enalaprilat is an option, but 10 mg every 6 hours would not be an appropriate starting dose because it would put the patient at risk of overshooting and, given the long duration of activity, could lead to unwanted consequences (Answer D is incorrect).

8. **Answer: A**

The initial goal reduction for this patient's BP, given that she is experiencing a hypertensive emergency, is a 25% reduction in MAP within the first 60 minutes (Answer A is correct). More rapid BP reductions may result in a lack of cerebral perfusion; therefore, they are not recommended (Answer B is incorrect). The patient is not experiencing any of the specific hypertensive emergencies (e.g., aortic dissection) that would call for a more rapid BP reduction. In addition, she is not experiencing a hypertensive emergency that would require a slower BP reduction (e.g., stroke) (Answers C and D are incorrect).